TOOL DESIGN

Subject Code : 23A03504a (Professional Elective - IV)

UNIT I : INTRODUCTION TO TOOL DESIGN

Syllabus (as given)

Introduction - Tool Engineering - Tool Classifications - Tool Design Objectives - Tool Design in manufacturing - Challenges and requirements - Standards in tool design - Tool drawings - Surface finish - Fits and Tolerances - Tooling Materials - Ferrous and Non ferrous Tooling Materials - Carbides, Ceramics and Diamond - Non metallic tool materials - Designing with relation to heat treatment.

1. Introduction to Tool Design

1.1 What is a tool?

- A tool is any device (cutter, die, jig, fixture, gauge) used to cut, form, shape, hold or inspect a workpiece in manufacturing.
- Tool design is the process of designing and developing the tools, methods and techniques needed to improve manufacturing efficiency and productivity.
- It is a specialised branch of production engineering - it joins the product drawing to the actual production on the shop floor.
- The person who does it is the tool designer ; the person who makes the tool is the tool maker .

Why tooling is needed

- To make parts interchangeable and of uniform quality
- To produce large quantities quickly (mass production)
- To reduce the skill needed from the operator
- To reduce cost per piece and inspection time

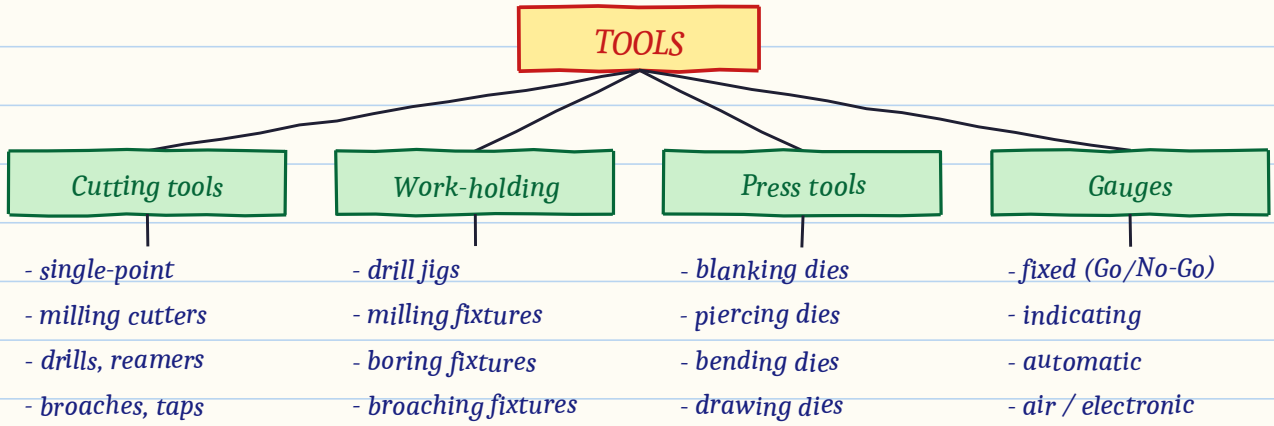
1.2 Tool Engineering

- Tool engineering is the branch of engineering concerned with planning, designing, making and managing all the tooling needed for production.

- It covers: tool design, tool planning, tool manufacture, tool inspection, tool control (store, maintenance, regrinding) .
- Functions of the tool engineer:
  - Study the part drawing and decide the process sequence (process planning).
  - Decide the machines, cutting tools, work-holding devices and gauges required.
  - Estimate the cost of tooling and the pay-back (economic justification).
  - Design the tools, prepare drawings and supervise manufacture and try-out.
  - Maintain, repair, recondition and reclaim the tooling.

1.3 Tool Classification

- Tools can be grouped by function as shown in the chart below.



Group	Purpose	Examples
Cutting tools	remove metal as chips	single point tool, milling cutter, drill, reamer, broach, tap
Work-holding	locate and clamp the work	drill jigs, milling/boring fixtures, chucks, vices
Press tools	cut or form sheet metal	blanking, piercing, bending, drawing dies
Gauges	check size, shape, position	plug gauge, ring gauge, snap gauge, indicating gauge
Special tools	assist assembly/forming	moulds, forging dies, welding fixtures, assembly tools

- Another classification: standard tools (available off-the-shelf, e.g. drill bits) and special tools (designed for one particular job, e.g. a jig for a casing).

- Based on production volume: general-purpose, semi-special, special-purpose (dedicated) tooling.

2. Tool Design Objectives

- To reduce the manufacturing cost per piece (lower labour and machining time).
- To increase the production rate by reducing set-up, loading and idle times.
- To maintain the quality and accuracy : parts must be within tolerance and interchangeable.
- To make the operation safe and easy - reduce operator fatigue, guard moving parts.
- To utilise machine tools fully and allow semi-skilled labour to do skilled work.
- To design tools that are simple, rigid, light, durable and easy to maintain / repair .
- To use standard parts wherever possible (bushes, screws, pins, clamps) to cut cost and lead time.
- To allow the use of less costly material or process by clever tooling.

Tool design = Economy + Accuracy + Safety + Productivity

- Design must balance tool cost against number of parts to be produced
- Break-even quantity: N = (tool cost) / (saving per piece)

3. Tool Design in Manufacturing

3.1 General tool design procedure

1. Statement of the problem - study part drawing, material, quantity, tolerances, machine available.
2. Analysis of the problem - operations needed, locating surfaces, clamping, forces involved.
3. Analysis of the requirements - accuracy, loading/unloading, chip disposal, cost limit.
4. Development of possible solutions - make several rough sketches (ideas).
5. Evaluation of the ideas - compare on cost, accuracy, time, safety; pick the best.
6. Design and detail - calculate forces, select materials, prepare assembly and detail drawings.
7. Prototype / try-out and testing , then modification.
8. Final drawings, bill of materials, release to the tool room and documentation.

3.2 Place of tool design in the manufacturing cycle

- Product design -> Process planning -> Tool design -> Tool manufacture -> Production -> Inspection.
- Tooling is a link between the product designer's drawing and the finished part.
- Good tooling affects: lead time, cost, quality, scrap, safety and flexibility.

3.3 Factors affecting tool design

- Part: size, shape, material, tolerance, surface finish, lot size.
- Machine: type, size, power, speed, table size, accuracy, available.
- Production: rate required, manual / automatic, shop skill, tool-room capacity.
- Economic: cost of tool, life of tool, delivery time, pay-back period.

4. Challenges and Requirements

4.1 Challenges in modern tool design

- Shorter product life cycles -> tooling must be ready faster (rapid tooling, CAD/CAM).
- Tight tolerances and better finish demanded by customers.
- New hard-to-machine materials (titanium, Inconel, composites) need new tool geometry and materials.
- High speed machining causes heat, vibration and tool wear problems.
- Flexibility : small batches and many variants -> modular and quick-change tooling.
- Cost pressure and need for lean manufacturing, less scrap.
- Skilled tool makers are scarce; need automation, CNC, EDM, 3D printing of tooling.
- Environmental rules: cutting fluids, dry machining, recycling of tools.

4.2 Basic requirements of a good tool

- Sufficient strength and rigidity to resist forces without deflection.
- Accuracy and repeatability of location of every workpiece.
- Wear resistance and hardness at the working edges; toughness against shock.
- Easy and quick loading and unloading, with fool-proof design (the part can be placed only one way).
- Provision for chip clearance and removal; coolant access.
- Low cost , interchangeable standard parts, easy maintenance and repair.
- Adequate safety - no sharp corners, guards, clamp that cannot slip.
- Good appearance and finish , proper marking and ease of handling (weight, handles, lifting hooks).

5. Standards in Tool Design

- Standardisation = using agreed sizes, shapes, materials and practices so that parts can be bought, replaced and interchanged.
- Advantages: lower cost, shorter lead time, easy replacement, better quality, fewer varieties in stores.
- Standard bodies: BIS (IS) - India, ISO - international, ANSI/ASME - USA, DIN - Germany, BS - Britain, JIS - Japan.

Standard	Used for
IS 696 / SP 46	engineering drawing practice (lines, views, dimensioning)
IS 919 / ISO 286	limits and fits for shafts and holes
IS 3073	assessment of surface texture (Ra values)
IS 2362 / ISO 2768	general tolerances (no tolerance stated)
IS 3403 / ISO	standard drill bushes, press tool parts, die sets

- Standard components in tooling: drill bushes, locating pins, clamps, T-bolts, nuts, springs, punches, die sets, dowel pins, screws .
- Always choose a standard part first; design a special part only if no standard one works.

6. Tool Drawings

- A tool drawing is the "language" by which the designer tells the tool maker how to make and assemble the tool.
- Types of tool drawings:
  - Layout / concept sketch: freehand first idea.
  - Assembly drawing: shows all parts in position with a bill of materials (BOM) and balloons.
  - Detail (part) drawing: each part separately with full dimensions, tolerances, material, heat treatment, finish.
  - Sub-assembly drawing for complex tools.

Items on a tool drawing

- Title block: tool name, part no., scale, material, drawn / checked by, date
- Views: front, top, side, sections - as few as are enough to show the shape
- Dimensions with tolerances , surface finish symbols, fits
- Notes: heat treatment (e.g. harden 58-62 HRC), grind all over, deburr, marking
- Bill of materials: item no., description, qty, material, standard / purchased
- Part drawing of the workpiece shown in chain lines for reference (in jigs/fixtures)

- Rules to follow:

1. Use BIS / ISO conventions - first angle projection (India), standard line types, scale 1:1 where possible.
2. Show only the necessary tolerances - tight tolerance costs more.
3. Give dimensions from datum (locating) surfaces to avoid tolerance build-up.
4. Mention material and hardness for every part.
5. Leave clear space for clamping, chip removal and operator hands .

7. Surface Finish

- Surface finish (roughness) is the fine irregularity left on the surface by the manufacturing process.
- Measured by Ra (CLA) = arithmetic average of the heights of peaks and valleys from the mean line, in micrometre (um).
- Ra = (1/L) x integral of |y| dx, over the sampling length L.
- Finish affects wear, friction, fatigue strength, corrosion, appearance, sealing and fit quality.

Process	Ra (um)	Typical tooling use
Rough turning / milling	3.2 - 12.5	non-working surfaces of bodies
Finish turning / milling	0.8 - 3.2	base of jigs, fixture bodies
Surface grinding	0.4 - 1.6	locating faces, die blocks
Cylindrical grinding	0.2 - 0.8	pins, punches, bush bore
Honing / lapping	0.05 - 0.4	gauges, bush holes, dies
Polishing	0.025 - 0.2	forming and drawing die surfaces

- Rule: the finer the finish, the higher the cost - so specify only what the function needs.
- Roughness symbol: a check mark with the Ra value written on it; N-grade (N1 to N12)

also used (N6 = Ra 0.8 um).

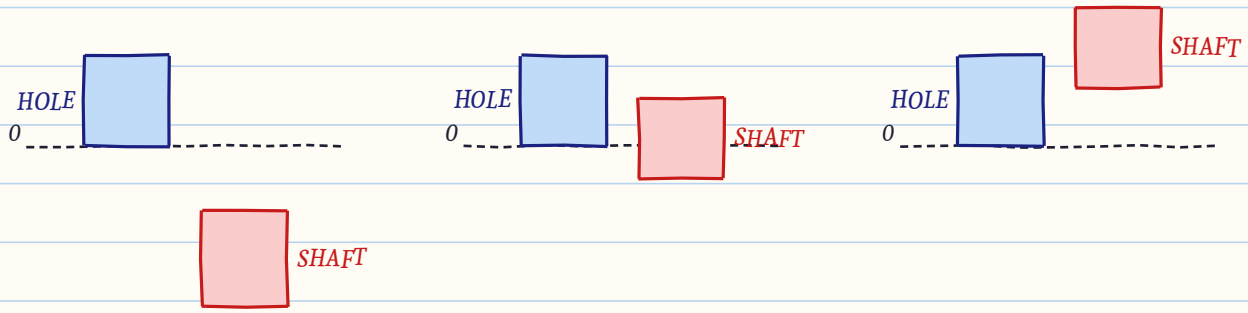
- Typical specifications in tools: locating surfaces Ra 0.8 , punch/die working surfaces Ra 0.4 or better , drill-bush bore Ra 0.4 .

8. Fits and Tolerances

8.1 Basic terms (IS 919 / ISO 286)

- Basic size: nominal size from which limits are measured.
- Limits of size: maximum and minimum permissible sizes.
- Tolerance = upper limit - lower limit (total permitted variation).
- Deviation: difference between actual/limit size and basic size. Fundamental deviation fixes the position of the tolerance zone.
- Allowance: intentional difference between mating parts (minimum clearance or maximum interference).
- Clearance = hole size - shaft size (positive). Interference = shaft - hole (positive).
- Unilateral tolerance: both deviations on one side of basic size. Bilateral: on both sides (e.g. 25 +/- 0.02).

8.2 Types of fits



Clearance: shaft always smaller | Transition: may be either | Interference: shaft always larger

Basic hole system: hole lower deviation = 0 (H), e.g. H7/g6 (clearance), H7/k6 (transition), H7/p6 (interference)

Fit	Condition	Tooling example
Clearance (running/sliding)	hole always larger than shaft	punch in guide, bush & liner sliding, H7/g6
Transition (location)	clearance or interference	dowel pin position, locating pin, H7/k6, H7/n6

Fit	Condition	Tooling example
Interference (press)	shaft always larger than hole	bush pressed into jig plate, liner, H7/p6, H7/r6

8.3 Systems of fits

- Hole basis system: hole size fixed (H, lower deviation = 0); shaft size varied to get the fit. Preferred - holes are made by standard drills/reamers, so fewer tools.
- Shaft basis system: shaft fixed (h, upper deviation = 0); hole varied. Used where one shaft carries many parts (e.g. line shafts).
- Tolerance grades IT01 to IT16: smaller number = finer tolerance. Gauges IT1-IT5, fits IT5-IT12, rough work IT12-IT16.

Typical fits in tool design

- Bush press fit in jig plate: H7/p6 (or H7/n6)
- Renewable (slip) bush in liner: H7/g6 or F7/h6
- Drill running fit in bush bore: F7
- Dowel pin in tool body: H7/m6, H7/n6
- Punch in stripper guide: H7/h6 (sliding)
- Punch and die: not a standard fit - clearance fixed by material and thickness

- Tolerance stack-up: total tolerance on a chain of dimensions = sum of individual tolerances -> keep chains short.
- Rule of thumb: tool tolerance should be about 10 - 20 % of the part tolerance (a tool 5 to 10 times more accurate than the part).

9. Tooling Materials

- Tool materials should have: hardness, wear resistance, toughness, hot hardness (red hardness), shock resistance, machinability and grindability, low distortion in heat treatment, availability and low cost.

9.1 Ferrous tooling materials

Material	Composition / Properties	Used for
Grey cast iron	C 3-3.5 %; good damping, casting, cheap	jig and fixture bodies, die shoes
Mild steel (IS 1570: 10C4, 20C8)	low C; easy machine, weld; low hardness	plates, bases, clamps; carburised parts

Material	Composition / Properties	Used for
Medium carbon steel (C40, C45)	0.4-0.5 % C; tough; hardenable moderate	bodies, shanks, shafts, pins
Plain carbon tool steel (T55 - T130)	0.6-1.3 % C; hard after quench; poor hot hardness (<200 C)	taps, files, punches, hand tools, simple dies
Case-hardening steels (15Ni..., En-36)	tough core, hard case	locators, bushes, pins, jig plates
Alloy tool steel (cold work: D2, O1)	Cr, Mn, W, Mo give deep hardening, low distortion	blanking dies, punches, gauges
Hot work tool steel (H11, H13)	Cr-Mo-V; hot hardness to 550 C	forging dies, extrusion, die casting
High speed steel (HSS)	18-4-1 (T1) or M2, M42 (Co); red hardness 600 C	drills, milling cutters, reamers, broaches
Stainless steel	Cr 12-18 %; corrosion resistant	food / medical tooling

- *HSS composition:* T1 = 18 % W, 4 % Cr, 1 % V; M2 = 6W-5Mo-4Cr-2V; adding 5-10 % Co (M42) gives higher hot hardness.
- *Ranking by hot hardness:* carbon tool steel < HSS < cast alloy < carbide < ceramic < CBN < diamond.

9.2 Non-ferrous tooling materials

- *Aluminium alloys:* light weight, easy to machine (use for large fixtures, hand-held jigs, prototypes); low wear resistance.
- *Brass and bronze:* good bearing material, non-magnetic, non-sparking; used for bushes, wear strips, liners.
- *Phosphor bronze, gunmetal:* bearings, bushes and worm-wheels in tooling.
- *Copper alloys (Be-Cu):* electrodes for EDM, spot welding tips, non-sparking tools.
- *Zinc alloys (Kirksite):* low-cost short-run drawing dies and moulds; easy casting.
- *Cast non-ferrous alloys / Stellite* (Co-Cr-W): tough, hot hard 800 C, cutting tips, wear faces.
- *Lead and low melting alloys:* used as soft jaws or to hold thin parts (pouring around the part).
- *Titanium / Tungsten / Molybdenum:* special high temperature tools.

9.3 Carbides, Ceramics and Diamond

- Cemented carbides: WC (tungsten carbide) grains bonded with Co by powder metallurgy (sintering).
  - Very hard (HRA 89-93), keeps hardness to ~ 1000 C; speeds 3-5 times HSS.
  - Grades: P (steel cutting; WC + TiC + TaC), M (multi-purpose), K (cast iron, non-ferrous; plain WC-Co).
  - More cobalt = tougher but softer. Brittle - needs rigid set-up. Used as brazed tips or indexable inserts , also coated (TiN, TiC, Al2O3).
  - In dies: carbide inserts for long-run blanking, drawing dies, bush liners, gauge wear faces.
- Ceramics: Al2O3 (alumina) with some ZrO2, or Si3N4, SiAlON.
  - Hardness to 1200 C, chemically inert; very high speeds on cast iron, hardened steel; but low toughness - need rigid, no interrupted cuts.
- Cubic Boron Nitride (CBN): hardness next to diamond; used for hardened steels (>45 HRC), stable to 1300 C and does not react with iron.
- Diamond (natural, synthetic PCD):
  - Hardest known material; cuts aluminium, copper, plastics, composites, graphite, glass with excellent finish.
  - Not used on ferrous metals - carbon diffuses into iron at high temperature; oxidises above ~ 700 C.
  - Also used for dressing grinding wheels, wire-drawing dies, lapping .

9.4 Non-metallic tooling materials

- Plastics: nylon, Teflon (PTFE), acetal (Delrin), bakelite, epoxy, fibre-glass - used for soft jaws, locator pads, wear pads, liners, light jigs ; they do not mar the finished surface.
- Epoxy / resin tooling: cast epoxy with filler (aluminium powder) for prototype dies, drilling jigs, check fixtures ; quick and cheap.
- Rubber and polyurethane: strippers, pressure pads, forming dies (Guerin process, rubber-pad forming), springs and cushions.
- Wood and laminates: patterns, templates, cheap fixtures for small quantity.
- Granite and ceramics: surface plates, bases of measuring tools (stable, no rust, no distortion).
- Cermets, graphite: EDM electrodes, hot forming die inserts.
- Advantages: light, cheap, corrosion-free, easy to make, good damping; disadvantage: low

strength and wear resistance, dimensional instability with moisture or temperature.

10. Designing with Relation to Heat Treatment

- Heat treatment changes the properties (hardness, toughness, grain) of steel by heating and cooling. Tool components are normally hardened and tempered .

Process	What is done	Purpose
Annealing	heat above critical temp, furnace cool	soften, relieve stress, improve machinability
Normalising	heat above critical, air cool	refine grain, uniform structure
Hardening	heat to austenite, quench in oil/water/air	maximum hardness
Tempering	reheat to 150-650 C after quenching	reduce brittleness and stress, adjust hardness
Case hardening	carburise / nitride / cyanide the surface	hard wear-resistant skin, tough core
Stress relieving	heat 500-650 C, slow cool	remove machining stress

10.1 Design rules to avoid heat treatment failure

- Distortion and cracking arise from uneven heating/cooling and sudden changes of section. So:

1. Keep the section thickness uniform ; avoid thick-thin joins.
2. Provide generous fillets and radii (min. 3 mm) at corners - sharp internal corners are crack starters (stress concentration).
3. Avoid sharp edges and notches ; chamfer or round the edges.
4. Avoid blind holes and holes near edges - quench liquid gets trapped (use through holes; plug with fire clay if blind).
5. Symmetrical shapes distort less; balance holes and slots.
6. Keep holes away from edges - at least hole diameter (min. 1.5 x dia.) of material around it.
7. Leave grinding allowance (0.2 to 0.5 mm) on hardened surfaces for finishing after heat treatment.
8. Where possible, use sectional (built-up) construction - a large complex die is made from small hardened blocks, easy to harden, grind and replace.

9. Rough machine -> stress relieve -> finish machine -> harden -> temper -> grind.
10. Choose the right steel : oil-hardening (O1) or air-hardening (A2, D2) for complex shapes (less distortion), water-hardening only for simple shapes.

Remember (exam points)

- Hardness of punches/dies: 58-62 HRC ; die inserts of D2: 60-62 HRC
- Jig bushes / locators: 58-62 HRC (case hardened, depth 0.8 - 1.5 mm)
- Springs and clamps: tempered to 40-50 HRC
- Jig bodies (cast iron / MS): normally not hardened but stress relieved
- Quench cracks: sharp corners, sudden thickness change, wrong quenchant

- Decarburisation (loss of carbon at the surface) is prevented by controlled atmosphere, vacuum or salt bath heating, or by leaving grinding allowance.

11. Quick Revision - Unit I

- Tool design -> link between product drawing and production; objectives = quality, cost, productivity, safety .
- Tool classes: cutting, work-holding (jigs-fixtures), press tools, gauges .
- Standards: IS, ISO, DIN, BS; use standard parts.
- Fits: H7/p6 press, H7/k6 transition, H7/g6 clearance ; hole-basis system preferred.
- Materials: C.I. (bodies), MS (plates), tool steels (dies), HSS (cutters), carbide / ceramic / CBN / diamond (hard machining).
- Heat treatment design: uniform sections, radii, no sharp corners, grinding allowance.

TOOL DESIGN

Subject Code : 23A03504a (Professional Elective - IV)

UNIT II : DESIGN OF CUTTING TOOLS

Syllabus (as given)

Mechanics of Metal cutting - Oblique and orthogonal cutting - Chip formation and shear angle - Single-point cutting tools - Milling cutters - Hole making cutting tools - Broaching Tools - Design of Form relieved and profile relieved cutters - Design of gear and thread milling cutters.

1. Mechanics of Metal Cutting

- Metal cutting = removal of unwanted material from a workpiece in the form of chips using a wedge-shaped, harder tool.
- The tool moves relative to the work with cutting speed V, feed f and depth of cut d.
- Cutting action: the tool compresses the metal ahead of it -> metal is sheared along a narrow zone (shear plane) -> the chip slides up the rake face of the tool.
- Three deformation zones: primary shear zone (work -> chip), secondary zone (chip-tool friction) and tertiary zone (flank-work rubbing).
- Heat generated: about 75 % goes with the chip, 20 % into the tool and work, rest in the surroundings.

1.1 Orthogonal and Oblique cutting

Feature	Orthogonal (2-D) cutting	Oblique (3-D) cutting
Cutting edge	perpendicular (90 deg) to cutting velocity	inclined at angle i (inclination) to the normal
Chip flow	chip flows over rake face, perpendicular to edge	chip flows sideways at an angle to the normal
Forces	2 components (Fc and Ft)	3 components (Fc, Ft, and radial Fr)
Edge length	edge length = width of cut	edge longer than width of cut -> more cutting edge in action

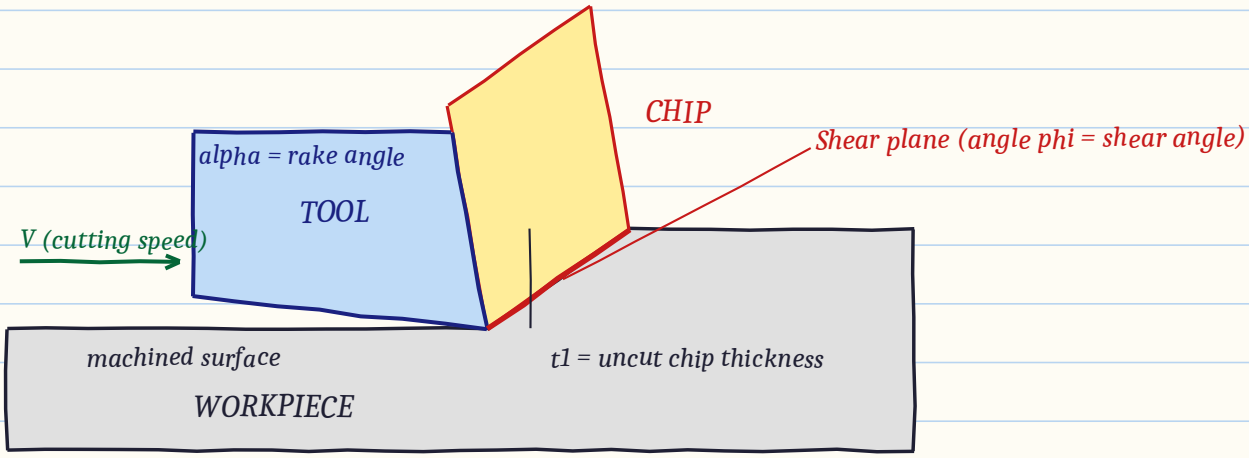
Feature	Orthogonal (2-D) cutting	Oblique (3-D) cutting
Use	theory, analysis, e.g. turning a tube end, broaching, planing	practical cutting: turning, milling, drilling
Surface	simple, easy to analyse	better finish and tool life (gradual engagement)

- Assumptions of orthogonal cutting theory (Merchant): sharp tool; 2-D deformation; continuous chip without built-up edge; uniform cutting velocity; shear occurs on one plane; width of chip = width of cut.

1.2 Chip formation and shear angle

$r = t1/t2 = \sin(\phi) / \cos(\phi - \alpha)$

$\tan(\phi) = r\cos(\alpha) / (1 - r\sin(\alpha))$



- Uncut chip thickness t1 (= feed in turning); chip thickness t2 after cutting; t2 > t1 always (chip is compressed and thickened).
- Chip thickness ratio (cutting ratio) $r = t1/t2 = \sin(\phi) / \cos(\phi - \alpha)$ ($r < 1$).
- Hence the shear angle $\tan(\phi) = r\cos(\alpha) / (1 - r\sin(\alpha))$.
- Chip compression ratio = $1/r = t2/t1$ (> 1). Larger shear angle -> thinner chip, lower forces, less heat -> better cutting .
- Shear strain in the chip: $\gamma = \cot(\phi) + \tan(\phi - \alpha) = \cos(\alpha) / (\sin(\phi)\cos(\phi - \alpha))$.
- Velocities: chip velocity $Vc = V\sin(\phi)/\cos(\phi - \alpha) = r\,V$; shear velocity $Vs = V\cos(\alpha)/\cos(\phi - \alpha)$.

Merchant's shear angle relations

- Merchant (min. energy): phi = 45 deg + alpha/2 - beta/2
- Lee and Shaffer (slip-line): phi = 45 deg + alpha - beta
- where alpha = rake angle, beta = friction angle (tan(beta) = mu = F/N)
- Positive rake or lower friction -> larger phi -> easier cutting

1.3 Types of chips

Chip type	How it forms	Conditions
Discontinuous (segmented)	metal fractures ahead of tool; small broken pieces	brittle work (cast iron, bronze), low speed, large feed, small rake
Continuous	ductile metal flows in a long ribbon; good finish	ductile metal, high speed, small feed, large rake, good lubricant
Continuous with built-up edge (BUE)	layer of work material welds to tool tip, breaks off periodically	ductile metal, low-medium speed, high friction; gives rough finish

- Chip breakers (groove or step on rake face) curl and break long continuous chips for safety.

1.4 Forces on the tool - Merchant's circle

MERCHANT FORCE CIRCLE

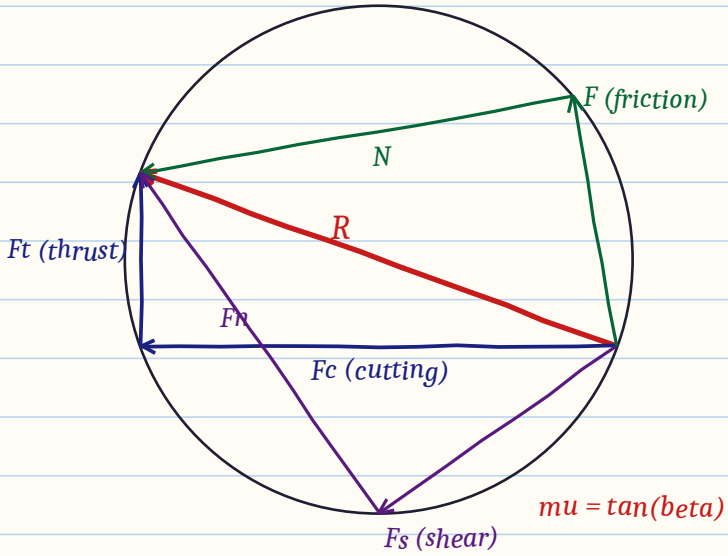
Diameter of circle = R

beta = friction angle = tan-1(F/N)

F = R sin(beta)
N = R cos(beta)

Fs = Fc cos(phi) - Ft sin(phi)
Fn = Fc sin(phi) + Ft cos(phi)

F = Fc sin(alpha) + Ft cos(alpha)
N = Fc cos(alpha) - Ft sin(alpha)



- Merchant's circle is a graphical tool to find all force components from just two measured forces (Fc and Ft from a dynamometer).
- Resultant R = sqrt(Fc2 + Ft2). Friction angle beta = alpha + tan-1(Ft/Fc).

- Friction: F = Fc sin(alpha) + Ft cos(alpha); N = Fc cos(alpha) - Ft sin(alpha); mu = F/N.
- Shear: Fs = Fc cos(phi) - Ft sin(phi); Fn = Fc sin(phi) + Ft cos(phi).
- Shear force = shear stress x shear plane area: Fs = tau_s x (b t1 / sin(phi)).
- Power: P = Fc x V (watt). Specific cutting energy = Fc/(b t1) (N/mm2); about 1.5 to 3 times for steel than for aluminium.

Solved idea (Exam)

- Given: t1 = 0.25 mm, t2 = 0.75 mm, alpha = 10 deg
- r = 0.25/0.75 = 0.333; tan(phi) = 0.333 cos10 / (1 - 0.333 sin10) = 0.328/0.942 = 0.348
- phi = 19.2 deg. Chip is much thicker -> heavy deformation

2. Single Point Cutting Tools

- Single point tool: one main cutting edge, used on lathe, shaper, planer, boring bar, slotter.
- Parts: shank (held in tool post), face / rake face (chip slides), flank (end and side), nose , cutting edges (side and end), heel .

2.1 Tool signature (nomenclature)

- ASA (American) system: back rake - side rake - end relief - side relief - end cutting edge angle - side cutting edge angle - nose radius .
- Example signature: 8 - 14 - 6 - 6 - 6 - 15 - 1 mm (HSS tool for steel).

Angle	Function / effect
Back (top) rake	controls direction of chip flow along the tool axis; +ve reduces cutting force
Side rake	main influence on shearing; +ve reduces force and heat; too large weakens edge
End relief (front clearance)	prevents rubbing of flank with machined surface (5-10 deg)
Side relief	prevents flank rubbing with work feed direction (5-10 deg)
End cutting edge angle (ECEA)	avoids end edge rubbing; 5-30 deg
Side cutting edge angle (SCEA, lead)	distributes load on a longer edge; 15-30 deg; thins chip
Nose radius	strengthens tip; better finish (0.4-1.6 mm); too big -> chatter

- Rake angle guide: hard brittle materials -> small or negative rake (carbide cast iron); soft ductile (aluminium) -> large positive rake 15-30 deg; steel (HSS) 8-14 deg.

2.2 Design of a single-point tool (shank)

- Tool shank is treated as a cantilever beam loaded by the main cutting force Fc at the tip.
- Bending moment M = Fc x L (L = overhang). For a rectangular shank b x h (h in direction of force):
  - Section modulus Z = b h2 / 6; bending stress sigma = M/Z = 6 Fc L/(b h2) <= allowable.
  - Hence h = sqrt(6 Fc L / (b x sigma_allow)). Usually h/b = 1.25 to 1.6 (b = 0.75 h for good rigidity).
- Deflection check: delta = Fc L3 / (3 E I), I = b h3/12; keep delta small (< 0.025 mm for finishing).
- Fc is estimated from: Fc = Ks x (d x f); Ks = specific cutting pressure (steel 1800-2500 N/mm2, C.I. 1000-1300, Al 600-800).
- Overhang L should be not more than 1.5 times the shank height for rigidity.
- Standard shank sizes: 12x12, 16x16, 20x20, 25x25 (square); 12x16, 16x25 (rectangular) mm; HSS bit brazed or clamped.

Tool material and construction

- Solid HSS tool bit - small tools
- Brazed tip tool - carbide tip brazed on MS shank
- Indexable insert tool - clamped throw-away carbide insert (most economical; no regrind)
- Tip is positioned with top clamp, pin lock or screw-on methods

3. Milling Cutters

- Milling = removal of metal by a multi-tooth rotating cutter while the work feeds; each tooth takes an intermittent cut.
- Classification:

1. Plain (slab) milling cutter: cylindrical, teeth on periphery only; straight or helical teeth; flat surfaces.
2. Side and face cutter: teeth on periphery and one/both sides; slots, steps.
3. Slitting saw: thin (0.5-5 mm) for cut-off, narrow slots.

4. *End mills:* teeth on end and periphery; slots, pockets, profiles; solid or inserted.
5. *Face mill:* large diameter, inserts on face; heavy flat surfacing.
6. *Angle cutters (single/double):* dovetails, V-grooves, chamfers. *T-slot cutter* and *Woodruff keyseat cutter* .
7. *Form cutters:* convex, concave, corner-rounding, gear, thread cutters.

- Another classification: *profile-sharpened* (sharpened on land behind edge) and *form-relieved* (sharpened on rake face only, profile retained).

3.1 Nomenclature and design parameters

- *Parts:* body, tooth, land, face (rake), gash, fillet radius, root, back-off, keyway, bore.
- *Angles:* radial rake (0-15 deg), clearance (3-10 deg primary, secondary 10-15 deg), helix angle (20-45 deg), land width (0.8-2 mm).

Design item	Typical value / rule	Remark
Number of teeth z	z = (0.04 to 0.06) D to ~ 2 x sqrt(D)	fewer teeth for roughing (more chip space)
Diameter D	from machine arbor, depth of cut	D = arbor dia x 2.5 to 3 (minimum)
Helix angle	20-45 deg (slab) ; 10-20 (side)	smooth cut, axial thrust to be balanced
Rake angle	HSS: 10-15 deg steel, 0-5 C.I., 25 Al	carbide: 0 to -5 deg
Clearance	HSS: 4-10 deg	carbide 6-10 deg
Face width	equal to width of work + overrun	

- *Cutting force on milling:* power P = Q x Ks ; Q = metal removal rate = w x d x feed rate (mm3/min); feed per tooth ft = f / (z N).
- *Cutting speed V = pi D N/1000 (m/min)* ; typical HSS on steel 20-30, on aluminium 150-300 m/min; carbide 3-5 times more.
- *Up (conventional) milling:* cutter rotates against feed -> chip thin to thick; needs rigid clamp. *Down (climb) milling:* with feed -> thick to thin; good finish, needs backlash eliminator.

4. Hole Making Cutting Tools

4.1 Twist drill

- Parts: point, body (flutes, land, margin, web), shank (straight or Morse taper), tang.
- Geometry: point angle 118 deg (steel, general), 90-100 for C.I., 130-140 for hard materials / plastics 60-90; helix (rake) angle 24-32 deg; lip clearance 8-15 deg; chisel edge angle 120-135 deg; web thickness = 0.12-0.2 D (thickens toward shank).
- Forces: Torque T = Ks-based: T ~ K D2 f0.8 ; thrust ~ K D f0.8; about 50-60 % of thrust is due to the chisel edge -> web thinning .
- Feed per rev: f = 0.01-0.02 D (steel); speed 20-30 m/min (HSS).
- Design points: flute length > hole depth + clearance; ratio L/D up to 5 normal; deep-hole -> gun drill with coolant through hole.
- Defects: oversize (unequal lips), walk (poor chisel), burr; fixed by correct grinding and using bushes.

4.2 Reamers

- Reamer: multi-flute tool to finish a drilled hole to accurate size and fine finish (H7, Ra 0.8-1.6). Removal is small: 0.15-0.5 mm.
- Types: hand reamer (straight/spiral flutes, square shank), machine (chucking) reamer, taper reamer, expansion/adjustable reamer, shell reamer, floating holder reamer.
- Teeth number z = 1.5 sqrt(D) + (2 to 4), even number, irregular (unequal) spacing to avoid chatter; spiral left-hand flutes for through holes with keyways.
- Design: lead (chamfer) 45 deg at the start, bevel 1-3 deg taper at the tip, margin land 0.1-0.2 mm, rake 0-5 deg, relief 5-8 deg.

4.3 Boring tools, counterbores, countersinks

- Boring bar - single point tool in a bar to enlarge a hole; short and stiff (L/D < 4) to avoid chatter.
- Counterbore - enlarges the end of the hole for bolt heads; countersink - cone 60-120 deg for flat head screws; spot facer - flat seat.
- Taps cut internal threads: hand, machine, spiral-pointed; tap drill size = major dia - pitch.

5. Broaching Tools

- Broaching uses a long multi-tooth tool pulled or pushed over/through the work; the rise per tooth does the feed; finished in a single stroke .
- Internal broach (holes, splines, keyways) and surface broach (flat, contour); very high production rate and accuracy (IT7, Ra 0.8).
- Parts of a pull broach: pull end (shank), front pilot, roughing teeth, semi-finishing teeth, finishing teeth, rear pilot, follower.
- Teeth design:
  - Rise per tooth (RPT / step) 0.01-0.1 mm; roughing 0.05-0.1, finishing 0.01-0.02, finishing teeth have no rise (sizing).
  - Pitch p = (1.76 to 2.2) x sqrt(L) (L = length of cut in mm) - for chip space; at least 2-3 teeth in cut simultaneously: p < L/2.
  - Gullet (chip space): depth h ~ 0.4 p, root radius r = 0.25 p-ish; must hold the chip curl: chip volume x 3-5 (compression).
  - Rake 5-20 deg (steel 10-15); relief 1-3 deg; land 0.1-0.5 mm.
  - Total broach length = pull end + front pilot + (roughing + semi-finishing + finishing) + rear pilot.
- Broach force: Fb = Ks x (width of cut per tooth) x (RPT) x (teeth in cut). Check tensile strength at the smallest section (the first tooth root): stress = F/A <= allowable (HSS ~ 300-400 N/mm2 safe working).
- Chip breakers (nicks staggered on teeth) for wide cuts. Material: HSS (M2, M3), or carbide tipped for high production.

Advantages / limits of broaching

- Fast, accurate, good finish, no skilled operator
- Tool is costly, only for high volume; one broach for one shape; for through shapes only

6. Design of Form Relieved and Profile Relieved Cutters

6.1 Profile (ground) relieved cutters

- Teeth are relieved by grinding the land behind the cutting edge (clearance angle) e.g. plain, side & face, end mills.
- After resharpening, the profile of the cutter changes when the tooth is sharpened on the relief land . Suitable only for simple shapes (plain, helical edges), and for standard

cutters .

- Rake angle is generally taken zero or small ; sharpened by grinding the back of the tooth, so the diameter reduces slowly.

6.2 Form relieved cutters

- Used for complex profiles (gear cutter, thread milling cutters, form tools). The tooth back is relieved by a cam (Archimedean spiral) so that the profile is kept unchanged after resharpening on the face (rake face) only .
- The relief is produced by cutting the tooth back on the lathe with a relieving attachment: radius falls uniformly with angle - an Archimedean spiral with constant drop.

Cam relief (design equations)

- Pitch angle between teeth = 360/z; drop (cam rise) K = (pi D / z) x tan(alpha_c) (alpha_c = clearance angle, 10-12 deg for HSS)
- Cam lift K = (pi D / z) tan(alpha); standard rise adopted as the next higher cam rise (e.g. 1.5, 2, 3, 4, 5 mm ...)
- Reduction in diameter after full life = (back-off x number of regrinds)
- Radial depth of tooth H = profile depth + (0.5 to 1.5 mm) clearance + 1/2 K + fillet radius r; H and tooth pitch fix fillet radius (r = 0.25 H)
- Tooth thickness at root ~ 0.5 to 0.6 x pitch; body thickness D - 2H for the bore + keyway

- Relief angle on the sides of the profile: sides also need clearance; for profile slopes at angle theta to the axis, the side clearance angle alpha_s = tan-1(tan(alpha) x sin(theta)); minimum 1-2 deg is kept.
- Why Archimedean spiral? So that the radius reduces linearly with angle and the clearance angle stays the same after repeated grinding.

6.3 Design steps for a form relieved cutter

1. Fix the profile to be produced and the machine arbor.
2. Select outside diameter D , bore d (arbor), no. of teeth z.
3. Select rake angle (0 for form cutters, so profile preserved), clearance (alpha = 8-12 deg).
4. Calculate cam rise K from alpha, z, D; check and adjust (Standard cam).
5. Decide tooth height H (profile depth + clearance + K + fillet) and fillet radius r.
6. Check that root diameter D - 2H > bore dia + (6 to 10 mm) for keyway and strength; adjust z and D.

7. Prepare drawing with all dimensions, tolerance, hardness 62-64 HRC (HSS).

7. Design of Gear Milling Cutters

- Gear teeth by form milling : a form-relieved (involute) cutter takes the shape of the tooth space; indexing head turns the gear blank after each space.
- The involute shape changes with number of teeth , so cutters are supplied in a set of 8 cutters (No. 1 to 8) for a given module and pressure angle; each cutter is exact for the smallest tooth number in its range, slightly thin tooth space for others.

Cutter no.	Teeth range	Cutter no.	Teeth range
1	135 to rack	5	21 to 25
2	55 to 134	6	17 to 20
3	35 to 54	7	14 to 16
4	26 to 34	8	12 to 13

- Design data (Module m, pressure angle 20 deg): addendum = m; dedendum = 1.25 m; tooth depth h = 2.25 m; circular pitch p = pi m; space width at pitch circle = pi m/2; pitch circle dia = m z.
- Cutter dimension: cutter tooth thickness at pitch = (pi m/2) - small allowance; depth of cut = 2.25 m (cutter height > 2.25 m).
- Cutter outside dia chosen from standard (e.g. m = 1: 50 mm; m = 5: 100 mm...). Cutter has involute form for the smallest tooth number in range , using the base circle Rb = (m z/2) cos(phi).
- Involute profile coordinates : x = Rb (sin theta - theta cos theta), y = Rb(cos theta + theta sin theta) - generated point by point for the profile table.
- Alternatives: hob (continuous generating, better accuracy), gear shaper cutter , rack cutter .

8. Design of Thread Milling Cutters

- Thread milling cutters cut external / internal threads on a thread milling machine (single-pitch or multiple-pitch cutter).
- Types: (a) single-form cutter (one-tooth, like a disc with thread form; for large pitch, e.g. worms, acme, trapezoid) and (b) multiple-form (many-ring) cutter (several rings, as

wide as thread length; cuts the full thread in about 1.2 rev of the work).

- Principle: cutter rotates at high speed on a slightly inclined axis (helix angle of thread, lambda = tan-1(lead / pi d)); work rotates slowly and the cutter moves axially one lead per work rev.

Design data

- Lead L = pitch x starts; helix angle lambda: tan(lambda) = L/(pi d_m) (d_m = mean dia)
- Cutter inclination = lambda; cutter profile differs from the thread normal section (corrected for helix angle)
- Cutter outside dia ~ 60-100 mm (single), 20-60 for multi-ring; tooth form angle 60 deg (metric, ISO) or 29 deg (acme)
- Cutter is form relieved (back-off by cam) so that profile stays after face sharpening
- Number of teeth 10-16 per rev; rake 0 deg for form accuracy

- Profile correction : because the cutter axis is inclined, thread space measured normal to the helix is slightly different from the axial section; for large helix angle use a corrected profile .
- Thread milling gives high accuracy at high production for lead screws, worms; internal thread milling can be done by CNC thread mill (single point, circular interpolation) .

9. Tool Wear, Tool Life and Cutting Fluids

- Tool wear types: flank wear (on the clearance face; VB = 0.3 mm limit for HSS/carbide finishing), crater wear (on rake face by chip; depth KT), notch wear (at depth-of-cut line), nose wear , edge chipping and thermal cracking.
- Causes: abrasion, adhesion (welding and tear), diffusion (carbide at high temperature), oxidation, fatigue from interrupted cuts.
- Tool life criteria: flank wear VB = 0.3 mm (uniform) or 0.6 mm (non-uniform); crater depth; surface finish deterioration; force rise; failure of the edge.

Taylor's tool life equation

- V T^n = C (V = cutting speed m/min, T = tool life in min)
- n = 0.1-0.15 HSS; 0.2-0.4 carbide; 0.4-0.6 ceramic
- Extended: V T^n f^a d^b = C (feed and depth also reduce life)
- Speed has the largest effect: raise V by 20 % -> life is cut roughly by half (HSS)

- Example: HSS tool, V = 30 m/min gives T = 60 min. n = 0.125. Find T at V = 36 m/min.

(30/36)^(1/0.125) = (0.833)^8 = 0.233 -> T = 60 x 0.233 = 14 min .

- Economic tool life (min cost): T_opt = (1/n - 1) x (Tc + Cc/Co) ; Tc = tool change time, Cc = tool cost per edge, Co = machine + labour rate per minute.
- Factors improving tool life: correct geometry, proper material grade, rigid set-up, coolant, reduce speed, avoid interrupted cut, sharp edge and honing.

9.1 Cutting fluids

- Functions: cooling (carry away heat), lubrication (reduce friction), washing away chips, corrosion protection, better finish.
- Types: neat oils (low speed, heavy cut - broaching, tapping), soluble (emulsion) oils (general high speed; mostly cooling), synthetic, semi-synthetic, gases / air (dry machining, MQL).
- Not used in milling of cast iron (dry or air), and with ceramic tools (thermal shock).

10. Worked Examples

Example 1 - Single point tool shank

- A tool has Fc = 2400 N, overhang L = 40 mm, shank width b = 16 mm, allowable bending stress 160 N/mm2. Find the shank height h.
- h = sqrt(6 Fc L/(b sigma)) = sqrt(6 x 2400 x 40 /(16 x 160)) = sqrt(225) = 15 mm .
- Use standard 16 x 16 mm square shank (h = 16 > 15, so safe).

Example 2 - Turning power

- Turning steel with d = 3 mm, f = 0.25 mm/rev, Ks = 2000 N/mm2, V = 60 m/min.
- Fc = Ks x d x f = 2000 x 3 x 0.25 = 1500 N .
- Power P = Fc V/60 = 1500 x 60 / 60 = 1500 W = 1.5 kW at the tool; for machine efficiency 80 % -> motor power = 1.88 kW.

Example 3 - Plain milling cutter data

- Cut a 40 mm wide slab, 4 mm deep, in steel using D = 80 mm, z = 8, N = 100 rpm, table feed 120 mm/min.
- Cutting speed V = pi x 80 x 100/1000 = 25.1 m/min .
- Feed per tooth ft = 120 /(8 x 100) = 0.15 mm/tooth .
- Metal removal rate Q = w x d x feed = 40 x 4 x 120 = 19200 mm3/min = 320 mm3/s.
- Power at cutter = Q x Ks = 320 x 2000 N.mm/s = 640 000 N.mm/s = 640 W (Ks = 2000 N/mm2 for steel).

Example 4 - Broach pitch

- Length of cut L = 40 mm; rise per tooth 0.04 mm. Pitch p = (1.76 to 2.2) x sqrt(40) = 11.1 to 13.9 mm -> take p = 12 mm .
- Check at least 2 teeth in cut: L/p = 40/12 = 3.3 >= 2 -> OK.
- Number of roughing teeth = (total stock to be removed)/RPT = 1.2/0.04 = 30 teeth.

Example 5 - Cam rise of a form relieved cutter

- Cutter D = 75 mm, z = 10, clearance angle 10 deg.
- K = (pi D/z) tan(alpha) = (pi x 75/10) x tan 10 = 23.56 x 0.1763 = 4.15 mm -> choose standard cam rise 4 to 5 mm (take 4.5 mm, adjust relief angle accordingly).

Common viva questions

- Why is shear angle important? What affects it?
- Difference between form-relieved and profile-relieved cutters?
- Why does drill need web thinning? Why is reamer pitch irregular?
- Why are gear cutters supplied in sets of 8?
- Why must broach pitch be selected carefully? (chip space, 2-3 teeth in cut)

11. Quick Revision - Unit II

- Orthogonal: 2 forces; oblique: 3 forces. r = t1/t2 = sin(phi)/cos(phi-alpha).
- Merchant: phi = 45 + alpha/2 - beta/2. Merchant circle gives F, N, Fs, Fn from Fc, Ft.
- Single point tool: shank is a cantilever: h = sqrt(6 Fc L/(b sigma)).
- Milling cutters: plain, side-face, end mill, form; hole-making: drill, reamer, tap; broach: RPT, pitch = (1.76-2.2) sqrt(L).
- Form relieved cutters keep profile by cam relief (Archimedean spiral); gear cutters come in a set of 8; thread cutters inclined by helix angle.

TOOL DESIGN

Subject Code : 23A03504a (Professional Elective - IV)

UNIT III : DESIGN OF JIGS AND FIXTURES

Syllabus (as given)

Introduction - Fixed Gages - Gage Tolerances - selection of material for Gauges - Indicating Gages - Automatic gages - Principles of location - Locating methods and devices - Principles of clamping - Drill jigs - General considerations in the design of drill jigs - Drill bushings - Methods of construction - Types of Fixtures - Vice Fixtures - Milling Fixtures - Boring Fixtures - Broaching Fixtures.

1. Introduction

- *Jig:* a work-holding device that locates, holds and also guides the cutting tool (drill bush). It is generally not fixed to the table - e.g. drill jig, reaming jig, tapping jig.
- *Fixture:* a work-holding device that locates and holds the work but does not guide the tool ; it is bolted to the machine table ; the tool is set with a set block / feeler gauge - e.g. milling, turning, grinding, boring, broaching fixtures.

Point	Jig	Fixture
Function	locates, holds and guides tool	locates and holds only
Tool guiding	by drill bushes	set block and feeler gauge
Fixing	not bolted (hand-held or on table)	bolted/clamped to machine table
Used on	drilling, reaming, tapping machines	milling, shaping, planing, grinding, turning
Weight	light	heavy and rigid (absorb cutting forces)

- *Advantages of jigs and fixtures:* interchangeability and accuracy; no marking-out; faster production and lower cost per part; semi-skilled operator; reduced inspection; safe; can use machine to its full capacity.

- Elements: (1) body / base, (2) locating elements, (3) clamping elements, (4) tool guiding (jig) or setting (fixture) elements, (5) fasteners, feet, clamps.

2. Gauges - Fixed Gages

- Gauge: an inspection tool used to check whether a dimension lies within its limits - gives "pass or fail" (go / no-go) information, not the actual size.
- Fixed (limit) gauges: have fixed dimensions corresponding to the limits of size.

2.1 Taylor's principle

- GO gauge should check all dimensions of the feature at the same time (full-form gauge: complete length of engagement, full circular form).
- NO-GO gauge should check only one dimension at a time (point contact) so that a fault in any single dimension is detected.

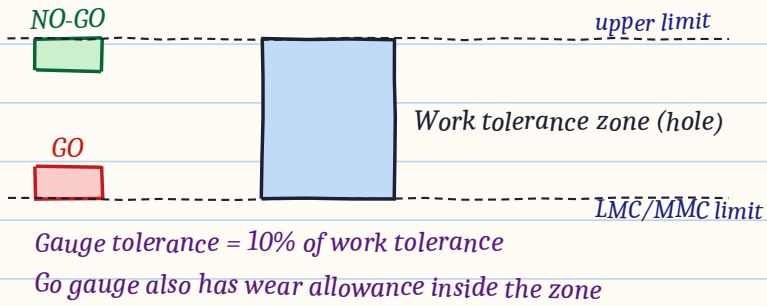
Types of fixed gauges

- Plug gauge (GO-NO GO) - for holes; double-ended or progressive; taper plug gauge
- Ring gauge and snap (gap/caliper) gauge - for shafts; adjustable snap gauge
- Thread plug / ring gauges, taper gauges, spline gauges
- Height, depth, flush-pin gauges , template and profile gauges, feeler (thickness) gauges
- Functional gauges / check fixtures - check position of holes, concentricity

- Working gauge (used by the operator in the shop), inspection gauge (used by inspector, slightly tighter), master (reference) gauge (to check the working gauges - the most accurate).
- Typical design details: GO plug is longer (full length), NO-GO is shorter and marked with a red band; handles (taper lock / trilock type); hardened, ground and lapped.

2.2 Gauge tolerances

- Gauges are themselves made to tolerances, which must be small compared to the work tolerance .



Gauge tolerance = 10% of work tolerance
Go gauge also has wear allowance inside the zone

Gauge tolerance (10% rule)

Work tol. = 0.05 mm
-> gauge tol. = 0.005 mm
Go gauge: tolerance zone placed inside the work limit (MMC side)
No-Go: placed to just stay within the work limit (LMC side)

- *Rule of 10 %:* gauge-maker's tolerance = 10 % of the work tolerance (for work tolerance of IT6 and above); this is split equally between GO and NO-GO gauges.
- *Wear allowance:* GO gauges wear in use; provide a wear allowance (about 5 % of work tolerance, or 0.0025 mm min) *inside the work limit* ; NO-GO gauges do not wear (rarely used) so no wear allowance.

Example

- Hole 25 H7: +0.021 / 0 mm -> work tolerance 0.021 mm
- Gauge tolerance = 10 % of 0.021 = 0.0021 mm; wear allowance = 5 % = 0.0011 mm
- GO plug: 25.0011 to 25.0032 (zone sits inside the work limit by the wear allowance)
- NO-GO plug: 25.0189 to 25.0210 (zone ends exactly on the upper limit)

- Inspection gauges: tolerance 5 % of work tolerance; master gauges: 1/10 of working gauge tolerance (tolerance zone of 10 % of 10 %).

2.3 Selection of material for gauges

- *Requirements:* hardness (>= 62 HRC), wear resistance, dimension stability (no distortion/ageing), good machinability / lapping, corrosion resistance, low thermal expansion, low cost.

Material	Remarks
High carbon high chromium (D2) / oil hardening steel	common, 60-64 HRC; stabilised and lapped
Case hardened MS (EN-8/ 15C8)	large gauges, low cost, case depth 0.8-1.2 mm; wear surface hard, core tough
Chromium plated steel	chrome layer 0.02 mm gives wear resistance and can be re-plated to restore size

Material	Remarks
Tungsten carbide (inserts)	very long life (30-50 times steel) for high volume work
Ceramic / glass / invar	for masters; very stable; low expansion
Aluminium / plastic	only for large light gauges (handles)

- Stabilisation: after hardening, gauges are cold treated (-80 C) and aged / tempered to remove retained austenite and avoid dimension changes.

3. Indicating and Automatic Gages

3.1 Indicating gauges

- Indicating gauge shows the actual deviation of the dimension from the nominal size on a scale/dial - a comparator.
- Common types:

1. Dial indicator (dial gauge): rack and pinion amplification; least count 0.01 mm (0.001 mm for precision).
2. Mechanical comparator (Reed type, Johansson Mikrokator): reed or twisted strip gives high magnification (x 1000-5000).
3. Optical comparator (optimeter): tilting mirror + lens, magnification x 800.
4. Electrical / electronic (LVDT) comparators: LVDT output amplified and shown as digital reading; resolution to 0.1 um .
5. Pneumatic (air) gauges: back-pressure or flow of air from nozzles shows clearance between nozzle and work; non-contact , magnification x 5000-10000; good for bore diameter, taper and out-of-round.

- Uses: checking size, ovality, taper, runout, flatness; fitted on snap, bore and depth gauge bodies; set with slip gauges (gauge blocks) or master pieces.
- Advantages: gives magnitude of error (statistical control possible); one gauge for many sizes; less operator skill. Disadvantages: costlier, slower, needs calibration, delicate.

3.2 Automatic gauges

- Automatic gauging: parts are gauged automatically on the machine or on the line and sorted or the machine corrected without the operator.
- Types:
  - Post-process (sorting) gauges: every part passes through gauge heads which classify it into 'oversize / good / undersize' bins by signals, e.g. ball-bearing sorting.

  - In-process gauges: measure while the part is machined (e.g. grinding): the signal stops the feed when the size is reached.
  - Post-process feedback gauges: gauge controls the offset/tool compensation of the machine (SPC loop).
- Sensors used: electronic LVDT heads, air-electric converters, capacitive and laser/optical sensors, vision systems , touch probes on CNC machines.
- Advantages: 100 % inspection at production speed, no human error, process control and statistics, lower scrap; limitation: costly, needs maintenance and calibration, justified for high volumes.

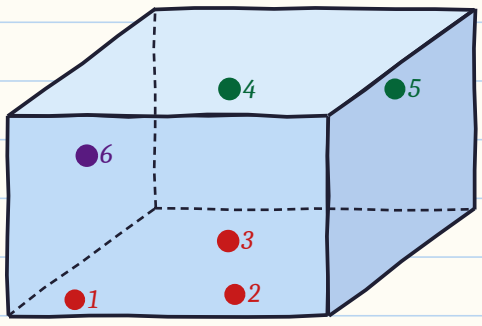
Gauges: quick compare

- Fixed gauge: fast, cheap, only pass/fail
- Indicating gauge: shows deviation, slow but precise
- Automatic gauge: fast + records + controls the machine

4. Principles of Location

4.1 Degrees of freedom

- A free body has 12 degrees of freedom : 6 axial (translation +/- along X, Y, Z -> 6 directions) and 6 radial (rotation clockwise + counter-clockwise about X, Y, Z).
- Locating means restricting some of these so that every workpiece takes the same position in the jig.



3-2-1 (six point) location principle

Points 1,2,3 : base (locating) plane
-> restrain 3 DOF (1 translation + 2 rotations)

Points 4,5 : guiding (back) plane
-> restrain 2 DOF (1 translation + 1 rotation)

Point 6 : stop (side) plane
-> restrain 1 DOF (1 translation)

Total = 6 DOF removed (3 + 3 rotational)

- 3-2-1 principle: locate with 3 points on the base (main) plane, 2 on the second (guide) plane, 1 on the third (stop) plane -> removes 9 of the 12 (the other 3 are removed by clamping force pressing the part against the locators; e.g. the 3 opposite directions).

- Redundant location (using more points than necessary, e.g. 4 points on base) causes the part to rock; use adjustable or equaliser pads where a casting surface is rough.
- Cylindrical part: a long V-block (or 2 short V-blocks) removes 4 DOF (2 translations + 2 rotations); an end stop removes 1 (axial); a key / flat peg stops rotation (1) -> total 6. Circular plate: 3 pads on the face, a central pin (2) and a side peg (1).

4.2 Choosing locating surfaces (datums)

- Select machined surfaces of the part whenever possible; if a casting surface, use 3 fixed points on the largest rough face (spherical head pins) or adjustable (screw jack) supports.
- Select locating surface close to the surface being machined ; the largest spread of the locators gives best stability and accuracy.
- Follow the drawing datum (design datum = locating datum) - this prevents tolerance stack-up and "locating error".
- Locators must be hard, wear resistant, replaceable , and easy to clean (chips can not collect).

5. Locating Methods and Devices

5.1 Locating from flat surfaces

- Fixed rest buttons / pads: flat or spherical headed - spherical buttons for rough cast faces, flat-headed pads for machined faces (dia 8-20 mm). Press-fit H7/p6, hardened 55-60 HRC.
- Adjustable supports (screw jack, spring-loaded): used for uneven castings, tall parts; locked after adjusting.
- Equalising (levelling) supports: a pivoted rocker that touches at more than 2 points; takes unevenness.
- Locating plates / rails; steps; nest (recess) matching outer profile of the part (a nest = a profile pocket, good for thin or small parts, with relief at corners and the depth about 1/2 of the thickness).
- Pins and diamond pins from existing holes (2 holes, one round + one diamond , avoids binding because of centre distance tolerance).

5.2 Locating from cylindrical surfaces (external)

- V-locators (V-block): 90 deg standard (60, 120 also used); centre the cylinder irrespective of diameter variation; the centre line shift vertical = delta d / (2 sin(V/2)); long V gives

4 point contact.

- Cup / cone locators, pad & plug: centralise the shaft end.
- Locating by nest / ring (sleeve), 3-jaw chuck, collet, expanding bush.

5.3 Locating from holes (internal)

- Plug (button) locators: round, with head ; clearance fit H7/g6 (slip) or press fit H7/p6; chamfer of the lead 1.5 mm; for accuracy fixed, hardened.
- Relieved (diamond) pin: the second pin has two flats cut away in the direction of the line joining the pins, leaving a narrow band (b ~ 3-6 mm) across it, so the variation in the centre distance of the two holes is absorbed and the part never binds.
- Expanding mandrels, tapered mandrel (1:2000 to 1:1500), cone locators, conical plug , and pull dowel .

Locating errors

- Error = clearance between locator and hole (max. hole - min. pin)
- Locating pins must be shorter than the hole depth and chamfered
- Hole location of 2 pins: centre distance tolerance of jig = (tolerance of part - clearance)/2
- Use sliding dowels ("pull") to compensate and an ejector for removing

5.4 Locating from profile and other methods

- Pin location from profile: two pins on a straight edge + 1 for the stop (3-2-1).
- Gap / fork locators; screw-type self-centring devices (concentric clamping: scroll, wedge, taper collet).
- Self-centring: all moving; mean position kept regardless of size, error reduced.
- Fool-proofing (poka-yoke): make the location asymmetric (extra pin or block) so the part can be inserted in the right way only.

6. Principles of Clamping

- Clamping is to hold the located workpiece firmly against the locators against cutting forces, weight and vibration without distorting it.

Rules of clamping

1. Clamping force should act on the locators (directed towards the locating surface, never away from them)
2. Clamp should press at rigid portions of the work , above a support , to avoid distortion
3. Force must be just enough to resist tool forces and not exceed; avoid marking finished surfaces (use soft pads)
4. Cutting forces must be taken by the locators and fixed jaws , not by the clamp
5. Should be quick, easy to operate - loading/unloading time is non-productive time
6. Self-releasing is avoided - clamp must not loosen with vibration (self-locking)
7. Clamp must be safe , not sharp, give clearance for the tool , chip and the operator's hands
8. Simple, strong, fool-proof , low cost, using standard parts (hooks, nuts, studs)

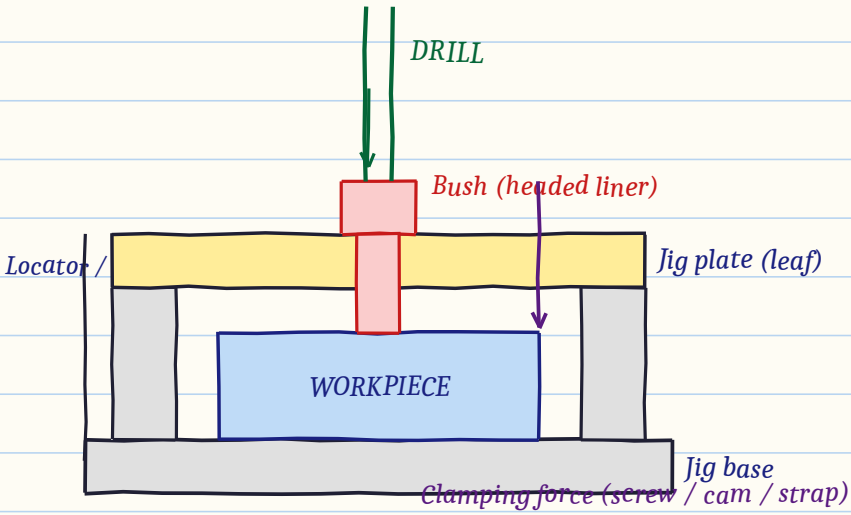
6.1 Types of clamps

Clamp	Working	Remarks
Strap (finger) clamp	bar pivoted over a heel; a nut/screw presses (lever of 1st, 2nd or 3rd class)	simple, cheap; for general use
Screw clamp	direct screw pressing the work; swivel pad	slow, high force
Cam clamp	eccentric or spiral cam on pin; rotating lever 90-180 deg	quick action ; self locking if angle < friction
Wedge clamp	taper wedge with 5 to 8 deg (self-locking)	quick; high force
Toggle (latch) clamp	links reach "dead-centre" so locked	very quick; low force; adjustable for thickness
Hook bolt / swing clamp	swings out of the way for loading	clear loading space
Equalising (floating) clamp	pivots to share force on 2-3 points	for uneven cast part
Hydraulic / pneumatic clamp	cylinder or diaphragm; pressure 4-7 bar	high production; automatic; CNC fixtures
Magnetic / vacuum chuck	holding by magnetism / suction	thin plates, grinding; no distortion

- Force relations: screw clamp: T = W x (r_m tan(alpha + phi) + mu_c x r_c), so W = T / (...) ; for a strap clamp with one end on work (L1) and bolt (L2) and heel (L3): clamp force on work = bolt force x L2/(L1 + L2) (bolt in the centre -> 50 %).
- Cam clamp: spiral curve rises uniformly; the rise angle must be less than the friction angle (about 5-8 deg) so that it self-locks; eccentric throw e = 1/2 (total rise) .
- Toggle: force at work = P / (2 tan(alpha)) - becomes very large near dead centre (alpha -> 0); used with an adjustable stop .

7. Drill Jigs

- A drill jig holds the part and guides the drill through a hardened bush to ensure position, perpendicularity and accuracy of holes, with no marking-out.



Parts of a drill jig

1. Jig body / plate
2. Locating elements
3. Clamping elements
4. Drill bushes (guide tool)
5. Feet (to rest on drill table)
6. Chip clearance & loading space

7.1 Types of drill jigs

Type	Features / use
Template jig	simplest: a plate with bushes laid on the work; for lightly built, large parts; low accuracy, low volume
Plate jig	plate with bushes + clamps and locators; accurate for flat parts; plate jig with feet clears chips
Open (table) jig	work is placed on a table; bush plate fixed; for parts drilled from one side
Leaf (hinged) jig	leaf hinged on a body, opens for loading; clamping by a thumb-screw/cam; small parts, drilling from one side

Type	Features / use
Box jig (tumble jig)	part enclosed, bush on several faces; box turned over to drill on different sides; heavy, rigid
Channel jig	U-shaped body, part fits into channel, clamped by screw or cam; simple; one or two sides
Angle plate jig	holes at right angle to the locating face; angle plate base
Indexing jig	drill several holes on a circle/straight line using index plate and plunger
Trunnion jig	large castings rotate on trunnions for drilling at several angles
Universal jig	adjustable/ modular for family of parts; with sub-plates
Pot jig / Diameter jig	cylindrical part held in a pot/V-block cradle; radial holes drilled
Pump jig	lever-operated (rack and pinion) quick lift of bush plate; for repetitive work

7.2 General considerations in the design of drill jigs

1. Study part drawing: material, size, quantity, operations, tolerance of holes and locating datum.
2. Provide clear access to load and unload the part with the leaf/ swing clamp; mistake-proof (one way only).
3. Clamp and locator must not interfere with tool path or the bush; the bush plate must be rigid ; thickness 1.5-2 x bush length to retain bushes (about 15-25 mm).
4. Hold the part against thrust of the drill - the clamp resists torque and the locator resists thrust : the cutting force goes into the base, not the clamp.
5. Provide chip clearance : a gap of 1 x drill dia (>= 1.5 mm for small) between bush end and workpiece; open at the bottom for chips; clearance slot / holes to remove chips; add coolant access.
6. Feet: 4 pad feet (or 3 for rock-free) at the base, ground flat; height >= drill extension, so that the drill's exit does not touch the table (also for through holes).
7. Keep jig light for handling (use cast aluminium for body) and easy to invert ; handles to lift; weight < 10 kg for hand use.
8. Rigid construction - fabricated welded, cast or built-up screwed and dowelled (two dowels each part).
9. Use standard components and hardened wear surfaces; provide identification marks ,

finishing, deburr.

10. Consider economics - cheapest design that meets accuracy; use jig only if number of parts justifies .
11. Fool-proof , safe, visible location of the work and inspection of locating surface cleanliness .

Tolerances in jigs

- Tolerance on tool dimensions = 20 - 50 % of the part tolerance (generally 1/5 to 1/3)
- Distance between bushes: part tolerance x 0.2 (+/- 0.02 for +/- 0.1 on the part)
- Bush axis perpendicularity: 0.01 / 25 mm

8. Drill Bushings

- Bushes guide the drill, fix its position and hold it against deflection. Made of hardened & ground tool steel (HSS / 60-64 HRC) or case-hardened steel; carbide for long run .

8.1 Types of drill bushes

Bush	Features
Press-fit bush (fixed): plain or headed	pressed in jig plate (H7/p6); for short run (not replaced); headed bush has flange to stop tool thrust
Renewable (replaceable) bush	used with a liner bush ; fixed by lock screw / clamp ; for long runs where wear needs replacement ; slip fit h6/G7
Slip (removable) bush	slip into liner; can be changed quickly for drill-ream-tap sequence on the same hole; with a lock screw or knurled/ latch
Screw bush	threaded into jig plate; adjustable for height; for clamping the part also
Special (large) bushes	Plate bush (rectangular), V-bush, angle bush , swivel / half bush for slanted surface , bush with extended nose for recess
Liner bush	a permanent bush press-fit in jig body to receive renewable bush

- Bush dimension (standard IS 3403): bore for drill: +0.010 to +0.025 (F7) above nominal size; bush length L = 1.5 to 2.5 x bush bore diameter (the longer, the better guidance); head dia = OD + 3 mm; the clearance between bush bottom and work = 0.3-1.0 x drill dia (for chip escape, but min. 1.5 mm).
- Bush bore: Ream bush (G6) , tap bush (H7) , drill bush (F7) : the bushes are given tolerances for the tool.

- Rule: material of the bush harder than tool shank ; surface finish Ra 0.4 in the bore; concentric to OD within 0.005 mm .

8.2 Methods of construction of jigs (jig bodies)

- Built-up (assembled) construction: parts of standard plate, bars, angles - screwed and dowelled together; cheap, quick; suits one-off.
- Welded (fabricated) construction: steel sections welded; light and quick; needs stress relieving & machining after.
- Cast construction: cast iron (grey) or cast steel / aluminium; good for complex shapes, damping & rigidity; pattern cost -> only for large quantity .
- Machined from solid / standardised modular systems: T-slot base plates and modular fixture kits (quick-change tooling) for flexibility.

Jig design (example check)

- Drill force (thrust) on a 10 mm drill in steel ~ 2000 N; torque ~ 12 N.m
- Locator and clamp must resist thrust (axial) and torque (rotation): clamp force x mu x radius >= T
- Add a safety factor of 2 to 3

9. Fixtures - Introduction and Types

- Fixtures are bolted to the machine table with tenon (key) blocks fitting in T-slots to align with the table; they are heavy & rigid to cope with large milling cutting forces and vibration .
- Setting: the cutter position is set with a set block (setting piece) and thickness (feeler) gauge of 2 to 3 mm to avoid cutting the set block (cutter never touches the fixture).
- Types of fixtures (by machine): milling, turning (lathe), boring, broaching, grinding, planing, shaping, welding, assembly, inspection fixtures; and by the power : manual, pneumatic, hydraulic.
- General design points: low centre of gravity , wide base, clamp over supports , chip clearance (large flushing holes), tenons , lifting lugs , wear plates, burr/cutter clearance .

9.1 Vice fixtures

- Use a standard machine vice with special soft jaws (shaped to the work) as the fixture : the vice acts as the clamp and base; cheap and quick.
- Types: plain vice, swivel vice, self-centring vice , hydraulic vice , tool-maker's vice , vice

with special inserts/V-jaws .

- Soft jaws (aluminium, mild steel, copper-alloy) machined to profile of part - jaw cavity cut in place for accuracy; replacable jaws .
- Design points: the part must rest on parallels or on a base; the movable jaw lifts slightly - use hold-down; the vice is aligned using a dial indicator or two tenons (keys) ; useful for small and medium parts, low cost .

9.2 Milling fixtures

- Used for slots, flats, key-ways, profiles, gear blanks . Must resist intermittent heavy cuts .
- Types: plain (single-part), string milling (a row of parts in a line, feed continuous), gang-milling (several cutters), straddle milling , indexing (rotary) fixtures, profile milling (cam/pattern controlled), rotary (continuous) fixtures (parts loaded on a rotating table or drum while cutting - reduces idle time to nearly zero).
- Design rules:

1. Clamp so that cutting force is directed on the fixed jaw/ locators (feed against the fixed locators ; cut in climb vs conventional).
2. Strong clamps - force at least 2x tool force; keep them close to the cut but out of the cutter path.
3. Rigid and heavy , low height, large base for stability; keep sections thick and avoid overhang .
4. Set blocks, tenons , ample chip clearance and flushing coolant , cutter overrun , and safe access for loading .
5. Fool-proof and lock-in so that the part cannot be loaded wrongly; wear plates and replaceable locators.

9.3 Boring fixtures

- Used on vertical / horizontal boring mills, lathes, jig-boring machines to bore holes of high accuracy (bearing bores, engine blocks, gear boxes).
- The tool (boring bar) is guided by bushes in the fixture (as in jigs) or the bar is supported at both ends in line with the spindle (to prevent deflection and chatter).
- Features: heavy base with tenons , locators ; clamps against cutting force ; bush (boring bar supports) with oil holes ; piloted boring bar with floating holder ; provisions for chips .
- Lathe boring fixtures: the fixture is mounted on the face plate (angle plate type) with counterweight to balance the rotating mass -> safety : no projecting parts, guards.

9.4 Broaching fixtures

- A broaching fixture holds the workpiece against a large pull / push force while the broach moves through - usually a simple plate (seating plate with a hole larger than the broach teeth) fixed on the broaching machine face .
- Types: (1) Fixed plate (face plate) with hole/ spherical seat : the part rests against the plate; the spherical seat floating (self-aligning) lets the part to align itself with the broach (e.g. for keyway or hole broaching). (2) Surface broaching fixtures with clamps , holding the part in a flat-bed and the cutting thrust is taken by the rigid body . (3) Rotary / indexing broaching fixtures for several surfaces (e.g. splined holes).
- Design points: the thrust (up to many tonnes) is taken on a strong seat ; clamp needed only for surface broaching; chips fall freely (slotted opening); guide the broach into the hole (pilot bush); load by hand easily ; fixtures should withstand shock and position stably .

Jigs vs fixtures - exam tips

- Jig -> bushes, light, drilling; Fixture -> set block, heavy, milling
- Locate by 3-2-1; clamp over locators; keep chip clearance
- Always mention fool-proof , standard parts , economy

10. Quick Revision - Unit III

- Taylor's principle: GO full form, NO-GO one dimension at a time. Gauge tolerance = 10 % of work tolerance.
- Indicating gauges: dial, mechanical, optical, electrical, pneumatic. Automatic gauges: in-process / post-process.
- 3-2-1 principle: 3 + 2 + 1 points; locating elements: pins, buttons, V-blocks, nests, mandrels.
- Clamps: strap, screw, cam, wedge, toggle, hydraulic; clamp force towards the locators.
- Drill jigs: template, plate, leaf, box, channel, indexing. Bushes: fixed, renewable, slip, liner; L = 1.5 - 2.5 d.
- Fixtures: vice, milling, boring, broaching - bolted to table; set block + feeler gauge.

TOOL DESIGN

Subject Code : 23A03504a (Professional Elective - IV)

UNIT IV : DESIGN OF PRESS TOOL DIES

Syllabus (as given)

Types of Dies - Method of Die operation - Clearance and cutting force calculations - Blanking and Piercing die design - Pilots - Strippers and pressure pads - Presswork materials - Centre of pressure - Strip layout - Short-run tooling for Piercing - Bending dies - Drawing dies - Design and drafting.

1. Introduction to Press Working

- *Press working* = shaping sheet metal (cold) by a *press* using a *punch and die set* - cutting or forming, in one stroke.
- Advantages: high production rate (10 to 1000+ strokes/min), low cost per piece, good accuracy and interchangeability, little scrap with good strip layout, semi-skilled labour.
- *Press tool (die set) parts:* upper shoe (punch holder), lower shoe (die shoe), punch, die block, stripper, guide pillars and bushes, shank, pilots, stops, back-up plates, springs, fasteners and dowels.

1.1 Press operations

Group	Operations
Cutting (shearing)	*blanking* (the cut-out piece is the part), *piercing / punching* (the hole is the part), notching, trimming, shaving, parting-off, slitting, lancing, cut-off
Forming	bending, drawing, flanging, curling, embossing, coining, stretching, bulging, extruding, forming (swaging)

- *Blanking* - slug is the product, sheet is scrap. *Piercing* - slug is scrap, sheet with holes is the product.
- *Lancing:* a partial cut and bending without removing metal. *Trimming:* cutting excess

from a drawn/formed part. Shaving: removing a thin layer to get a smooth accurate edge. Notching: cutting from the edge.

2. Types of Dies

2.1 Classification by operation

1. Simple (single-operation) die: one operation per stroke (blanking or piercing only).
2. Compound die: two or more cutting operations at the same station in a single stroke , e.g. blank + pierce a washer; the blank is cut upside down (inverted) - very accurate (concentric) but costly, slower.
3. Combination die: a cutting plus a forming operation at the same station in one stroke (e.g. blank & draw).
4. Progressive (follow-on) die: several operations at successive stations ; the strip advances one pitch per stroke; finished part at the last station . Highest productivity; pilots needed for accurate registering.
5. Transfer die: the part is cut first, then transferred mechanically from station to station for forming (for large parts, deep drawing).
6. Multiple (gang) die / sub-press die for small parts.

Type	Merits	Limitations
Simple	cheap, easy, short lead time	only one operation, low output
Compound	accurate, flat parts, single set-up	expensive, more difficult, slower than progressive
Progressive	high speed, automatic, one man for several ops	high tool cost, big strip needed, high skills to maintain
Transfer	large parts, deep draws	transfer mechanism costly

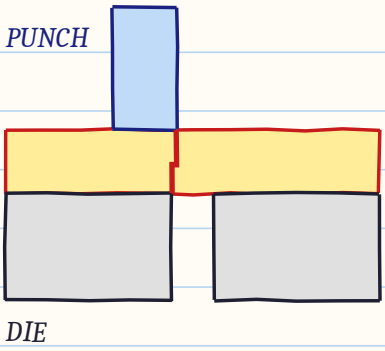
2.2 Classification by construction

- Inverted die: die block on the upper shoe, punch on the lower; the blank is pushed down by a knock-out (flat blanks, compound dies).
- Pillar (guide-post) die sets: two or four guide pillars and bushes keep alignment of punch and die (diagonal, back-post, centre-post). Plain (open) dies: no guide - the press ram guides (cheap, low accuracy).
- Sub-press die: a small die set with guide that is placed in the press; used for small precision parts.

3. Method of Die Operation

- Shearing occurs in three stages as the punch descends into the sheet on the die:

1. Plastic deformation - sheet bends and flows into the die opening; edge shows rollover .
2. Penetration (shearing) - punch cuts about 1/3 (for soft) to 1/4 of the thickness (hard materials); smooth burnish band formed.
3. Fracture - cracks start from the cutting edges of punch and die, meet, and the slug is separated; rough fractured zone and burr .

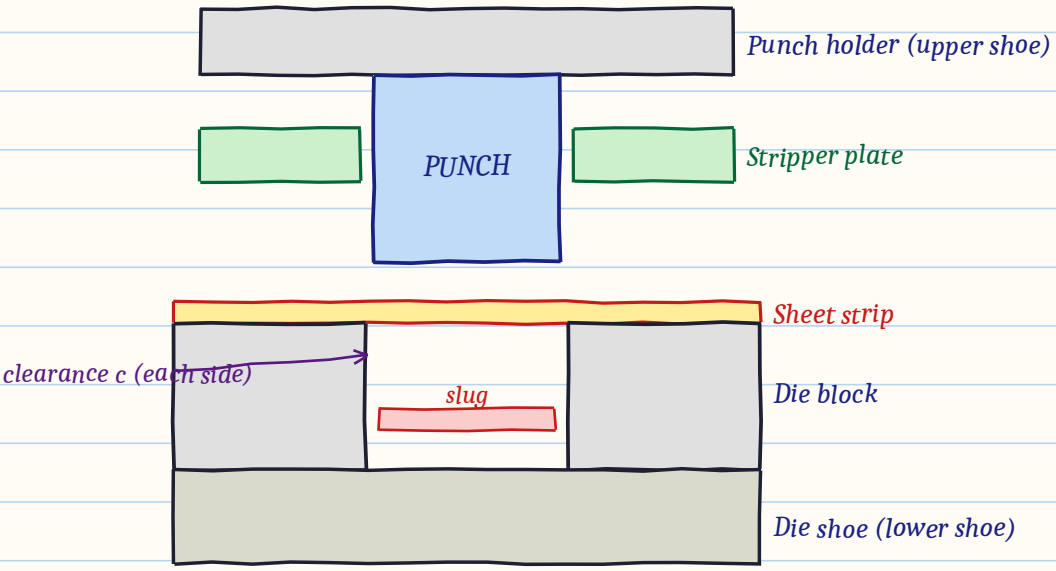


Cut edge of blank shows 4 zones:

1. Rollover (burnish rounding)
2. Burnish (clean sheared band)
3. Fracture zone (rough)
4. Burr (on die side of the blank)

Too little clearance -> secondary shear, high force
Too much clearance -> large burr / rollover

- With the correct clearance , the cracks meet, giving the minimum cutting force and a clean edge .
- Slug goes through the die (relief angle 0.25-2 deg or straight land 3-10 mm then relief of 1-2 deg per side).
- Die land (straight portion): 3-10 mm; after wear re-grind; angular clearance 0.5-2 deg per side.
- Strip is stripped by the stripper (spring or fixed) from the punch on the return stroke, since the sheet grips the punch after cutting (shrinks on the punch).



Blanking die: slug (blank) falls through die opening; strip scrap is stripped from punch

4. Clearance and Cutting Force

4.1 Clearance

- Die clearance c = gap between punch and die opening, per side . It depends on thickness t, material strength and quality of edge desired .

Clearance (per side)

- c = 0.0032 x t x sqrt(tau_s) (mm) ... tau_s = shear strength in N/mm2 (t in mm)
- Thumb rules: c = 5 % t (soft aluminium, brass), 7 - 10 % t (mild steel), 10 - 12 % t (hard steel, stainless)
- Thin sheet (t < 1 mm): c ~ 3-5 % t; Clearance on both sides = 2c

- Too little -> double shear (secondary burnish), higher force, faster punch wear , more stripping force.
- Too much -> large burr, rollover , tapered slug, less accuracy.

4.2 Punch and die sizes (applying clearance)

- Blanking: the blank size = die size (die opening made to blank size); punch = die opening - 2c (clearance is taken on the punch).
- Piercing: the hole size = punch size (punch made to hole size); die opening = punch + 2c (clearance taken on the die).
- Why? Cut edge is the part of the sheet that touches the sharper tool: for a blank it's the die edge; for a hole it's the punch edge.

- Allowance for wear: add (blank) or reduce (hole) by one fourth of tolerance (to start at the worst side of the tolerance) so that the tool wears into the limit.
- Tolerances on punch & die: 5-10 % of the part tolerance (e.g. IT6 / IT7).

4.3 Cutting force

Force formulas

- Cutting force: F = L x t x tau_s (L = length of cut edge = perimeter) [N]
- Shear strength tau_s ~ 0.7-0.8 x ultimate tensile strength
- Reduced with shear on punch/die (slope or taper): F_shear = F x (k t / s) ~ (0.3 to 0.6) F, shear s = 1 to 2 t
- Stripping force = 2.5 to 20 % of cutting force (average 10 %)
- Work done: W = F x t x k (k = 0.4 - 0.6 penetration factor); press power -> tonnage with 20-30 % extra margin
- Press capacity (tonnes) = 1.3 x F / 9810 (safety margin 30 %)

- Example: blank a washer OD 50 mm, ID 20 mm from t = 2 mm mild steel strip, tau_s = 300 N/mm2.
  - L = pi (50 + 20) = 220 mm; F = L t tau = 220 x 2 x 300 = 132 000 N = 13.5 tonne
  - Stripping force 10 % = 1.35 t; press capacity needed = 1.3 x 13.5 ~ 18 tonne -> choose 20 t press.
  - Clearance c = 0.0032 x 2 x sqrt(300) = 0.11 mm per side.
- Reducing cutting force: (1) provide shear (inclined cutting edge) on the punch or die; (2) stagger punch lengths in multiple-punch tools, (3) use a heavy flywheel press.
  - Shear on punch for piercing (die flat) and shear on die for blanking (the punch face flat) so as to keep the part flat . Shear = 1 - 2 t.

5. Blanking and Piercing Die Design

5.1 Die block (die plate)

- Material: tool steel (D2, D3, O1 or T105 / HCHCr), hardened 58-62 HRC; carbide inserts for very long runs; mild steel or cast iron blocks with inserts for large dies.
- Thickness H (thumb rule): H = 0.4 x (cube root of F in N), minimum 12-15 mm; usually 20-40 mm. Thin die for small force, thicker for thick sheets.
- Edge distance (wall): distance from the opening to the edge of the die block = 1.5 to 2 x H (min. 25-30 mm); distance between two openings >= 1.5 H.
- Screws and dowels: at least 2 dowels (diagonal) and 4 socket screws , placed 1.5 x dia.

from the edge - screws M8-M12 spaced about 60-100 mm; keep them clear of the opening .

- Die opening: straight land 3-10 mm then a relief of 0.5-2 deg per side (or 1.5 mm-step relief) so the slug falls free; for sheet thickness less than 1.5 mm, a land = 3 mm.
- Sectional dies (die in pieces) for large, complex shapes: ease of heat treatment, repair and grinding; join lines at corners avoided.

5.2 Punch design

- Material: HCHCr / D2 / HSS, 58-62 HRC; shank (head) softer (30-45 HRC).
- Length of punch: L = (stripper plate thickness + back-up + guiding + (1 to 1.5 t) penetration + 2-5 mm).
- Fixing: (a) head (shoulder) type with retaining plate; (b) screw (plain punch with counter-bored screws); (c) dowel / pins ; (d) shank held in a punch holder; (e) potted with low-melting alloy / epoxy .
- Buckling check of a slender piercing punch (hole dia d < 1.5 t, or d < t for thin punches):
  - Critical load (Euler) Pcr = pi2 E I / (k L)2 ; I = pi d4 / 64; end condition k = 0.5 to 1.
  - Safe if cutting force F = pi d t tau < Pcr / FOS. Piercing diameter d_min ~ 1.0 t (steel) to 0.8 t (soft) without a guide; use a guide (stripper bush) for small punches.
- Punch face: flat, or stepped (staggered) / sheared by 1-2 t to reduce the force and noise; piercing punches with a bulleted nose to hold the slug (slug retaining by a slight dimple/groove).
- Back-up plate (pressure plate): hardened 45-50 HRC; when the punch pressure on the punch-holder > 150-200 N/mm2 , to prevent the punch head sinking into the holder.

5.3 Pilots

- Pilots are small pins that locate the strip precisely in the die before the punch cuts, by entering already-pierced holes - in progressive dies (indirect pilots in the scrap) and compound dies .
- Types: direct pilot (enters a hole that is part of the product - small, 'bullet nose'), indirect pilot (enters a hole punched in the scrap web for registering only).
- Design points:
  - Pilot diameter 0.05 - 0.15 mm smaller than the pierced hole (the hole is placed with a precision of +/- 0.02-0.05 mm).
  - Pilot length = 1 to 1.5 t beyond stripper surface; nose is bullet / conical (taper 30 deg) to move the strip smoothly.

- Pilot must enter before the punch cuts (set 1-2 t longer than punches) and the strip must be free to float (lifted by lifters /guides).
- Pilot head is spring loaded (pressure) or fixed ; pilot is hardened (55-60 HRC) HSS ; replaceable, pilot hole size same as punch/die size set.

- Stock lifters and stops: finger stop, trigger stop (hand-held), automatic stop, side-cutter (pitch notch) , start (stock) stop for first piece, guide rails (strip guide gap = strip width + 0.1 to 0.5 mm).

6. Strippers and Pressure Pads

- Stripper: a plate that holds the strip down and strips it from the punch on the up stroke. It also guides small punches .

Type	Description	Use
Fixed (channel) stripper	a rigid plate fixed on the die with a slot for strip; gap = t + 0.1-0.3; height of slot = 1.5 t + 1 mm	simple; low cost; thin to medium strips; cannot hold work flat
Spring (movable) stripper	plate spring-loaded on the punch holder; presses on sheet before punch cuts	keeps the blank flat; cleaner edge; guides punches; for thin sheets
Positive (knock-out) stripper	rod-actuated or cam-actuated	where spring force is not enough
Pressure pad / knock-out	spring or rubber loaded pad in the die that ejects the blank (inverted dies)	flat blanks; compound dies
Pneumatic / rubber pad	compressed air or urethane	large dies, big forces

- Stripping force: F_s = 2.5 % - 20 % of cutting force (usually 10 %).
- Spring selection: force per spring = F_s / n; stroke of spring = t + penetration + grinding allowance (5-6 mm); use die springs (round wire, rectangular/ heavy) pre-loaded ; or polyurethane/ rubber (compressed 15-25 %).
- Rubber stripping pads: area x pressure; polyurethane pad of 1/3 stroke.
- Stripper plate thickness = 12-20 mm or 0.2 x the strip width , hardened (guided punches) or mild steel; distance from punch to hole in stripper = 0.05 - 0.1 mm (sliding fit) .

6.1 Knock-outs / ejectors

- In inverted and compound dies the blank stays in the die or around the punch -> knock-out (ejector pad) pushes it out by press knock-out bar and pins , spring or air .

7. Presswork Materials

- Press-work stocks should be ductile (high elongation), uniform and with good surface.

Material	Properties	Typical products
Low carbon steel (CRCA, deep drawing steel, IS 513 - D, DD, EDD)	ductile, cheap, good formability	automobile panels, brackets, washers, covers
Stainless steel (304, 316)	high strength, work-hardens	sinks, utensils, medical parts
Aluminium and alloys (1100, 3003, 5052)	light, soft, corrosion resistant	cans, utensils, aerospace, panels
Brass (70/30 cartridge brass), copper	very ductile; high conductivity	electrical terminals, cartridge cases, decoration
Spring steel / phosphor bronze / Be-Cu	high elastic limit	springs, clips, contacts
Non-metals: paper, fibre, leather, rubber, plastic films	cut with steel-rule (cutting) dies	gaskets, insulation, shims

- Stock supplied as sheet, strip or coil ; thickness tolerance +/- 5 % , hardness: dead soft, quarter-hard ... full-hard (bending needs soft, blanking prefers semi-hard for clean cut).
- Shear strength of mild steel ~ 300-350 N/mm2; aluminium 70-100; brass 220-300; stainless 450-500; copper 150-200 N/mm2.
- Grain direction (rolling direction): bends should be across the grain (perpendicular) to avoid cracks; blank orientation considered in strip layout.

8. Centre of Pressure

- *Centre of pressure:* the point where the resultant of all cutting forces acts - the press ram axis (shank axis) must pass through it ; otherwise eccentric loading -> tilting, uneven punch wear, ram guide wear, poor accuracy, breakage .
- *Calculation:* treat each cutting edge by its length (cutting force proportional to the length of the cut edge for the same thickness and material): take moments of the line lengths about two axes .

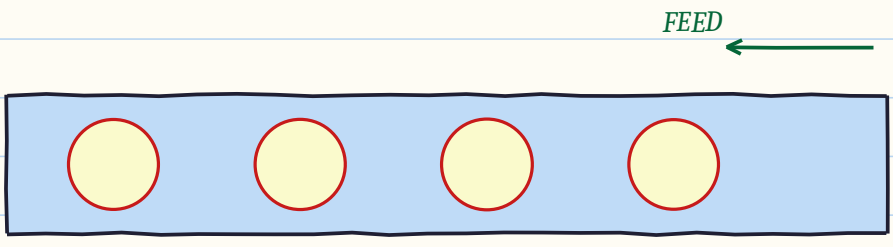
Formulae

- $x = (L1\ x1 + L2\ x2 + L3\ x3 + ...)/(L1 + L2 + L3 + ...)$
- $y = (L1\ y1 + L2\ y2 + L3\ y3 + ...)/(L1 + L2 + L3 + ...)$
- L = length of each cut (circle -> pi d, rectangle -> 2(a + b), line segments - their length); (x, y) its centre (centroid) from the chosen reference axes

- *Example:* piercing 2 holes: hole A dia 10 mm at (20, 20); hole B dia 20 mm at (60, 20); perimeter L1 = 31.4, L2 = 62.8.
  - x = (31.4 x 20 + 62.8 x 60)/(94.2) = (628 + 3768)/94.2 = 46.7 mm ; y = 20 mm . Hence the shank is located at (46.7, 20) - near the bigger hole.
- *Graphical method:* represent each force by a length and find the point with a funicular polygon ; also for non-symmetrical parts the shank is shifted . Put the die set so that the centre of pressure coincides with the shank .
- For progressive dies: use the total force at all stations in a stroke (including the stripping), and consider the forming forces also.

9. Strip Layout

- *Strip layout* = arrangement of the blanks on the strip so as to get maximum economy of material (minimum scrap) with adequate strength of the strip and die .



Strip layout

a = 0.8 t to 1.5 t (min. ~ 1.5 mm)
b ~ 1.5 t to 2 t
Strip width = D + 2b + clearance
Material utilisation = (blank area x blanks) / (strip area used) x 100

9.1 Design points

- Bridge (web) a between blanks and margin b on edges:

Thickness t	Web / margin
Up to 0.8 mm	a = 0.8 - 1.0 mm
0.8 - 3.2 mm	a = t to 1.2 t (min. 1.0 - 1.5 mm); b = 1.5 t
Over 3.2 mm	a = 0.8 t ; b = 1.2 t

- Strip width W = D + 2b + c (c = side clearance for guide 0.2 - 0.5 mm); pitch (feed) P = L + a (L = length of blank in feed direction).
- Material utilisation (%) = (area of blank x blanks per stroke) / (width x pitch) x 100. For good layout 70 - 85 %.

Example

- Circular blank D = 40 mm, t = 1.5 mm; a = 1.5 mm, b = 2.25 mm
- W = 40 + 2 x 2.25 = 44.5 mm; P = 40 + 1.5 = 41.5 mm
- Utilisation = (pi/4 x 40 x 40) / (44.5 x 41.5) = 1256.6/1846.8 = 68 %

- Layouts: single row, double row / staggered (blanks nested: better utilisation), angled (for L,T shapes), multiple row , and scrapless (parting, cut-off dies, no waste).
- Provide pilot holes in scrap , feed direction , grain direction , and burr side (burr should face away from the visible/ assembly side).
- In progressive dies, sequence of operations: pierce first (the small holes), notching, then forming/bending, finally blank (cut-off) ; idle stations to strengthen die; carrier strip held to the end.

10. Short-run Tooling for Piercing

- For low volumes (prototype, a few hundred to a few thousand pieces) - cheap, quick and flexible tooling is used instead of full die sets.
- Methods:

1. Pierce and blank dies made of cheap material: mild steel or Kirksite (zinc alloy) die with a plastic or low-hardness punch ; epoxy-faced dies with steel punch.
2. Dimple / sub-press dies: simple punch and die plate with a stripper using standard components (punch & die sets from catalogues), with locating gauge plate .
3. Rubber (urethane) pad (Guerin / Marform) piercing: a steel punch and soft rubber pad as a die ; works for thin soft sheet metal and gaskets.

4. Steel-rule (cutting) dies: steel knife edge bent to profile and set in wood/plywood block , rubber ejection pads ; for paper, leather, plastic, rubber, thin soft metals.
5. Turret punch presses (CNC) and nibbling: a standard-shaped punch in turret , no special tool ; the sheet moves on a CNC table ; laser/plasma cutting is also used instead.
6. Universal/ modular piercing units: pre-assembled punch holders on a base plate (like hole-punching units), quick set-up ; standard shape holes (round, square, slot) .
7. 3-D printed polymer or aluminium dies for very short runs.

- Advantages: low cost, quick delivery, easy changes; limits: low accuracy, low life, thin sheets only, slower.

11. Bending Dies

- Bending: plastic deformation of sheet about a straight axis ; outer fibres stretch, inner fibres compress ; the neutral axis shifts toward the inside.
- Types: V-bending (die), U-bending, wiping (edge) bending, air bending, rotary bending, bottoming, coining; channel, offset, curling, hemming, seaming.

11.1 Bend allowance and development

Bend allowance (length of neutral axis)

- BA = (pi / 180) x theta x (R + k t) (theta = bend angle in deg; R = inside radius; t = thickness)
- k = 0.33 (R < 2t), 0.4 (2t < R < 4t), 0.5 (R > 4t)
- Developed length = sum of flat lengths + sum of BA
- Minimum bend radius: soft steel 0.5 t, aluminium 1 t, hard brass 2 t, hard stainless 3 t (across grain) - not less, else cracks

- Springback: after the load is removed, the sheet partially recovers - angle opens by 1-10 deg. Springback is more in high strength material, large radius and thin sheet. Compensation: over-bending (die angle smaller than required by the springback angle), bottoming (coining), stretch bending , radius smaller , pressure pad .
- Springback factor: Ks = (R_i + t/2) / (R_f + t/2), where R_i = radius of the tool and R_f = radius after springback. Ks < 1, so the tool is made with a smaller radius / angle (overbend) to get the required final bend.

11.2 Bending force and die design

- V-die: F = (C x sigma_u x L x t2) / W (C = 1.33; W = die opening = 8 t; L = length of bend)
- U-die (channel): F = (0.67 to 0.8) x sigma_u x L x t2 / W ; wiping die: F = (0.33 sigma_u L

t2) / W, plus pressure pad force (30 - 50 % of F).

- Die design: die radius (entry) 3 - 5 t, punch radius = required inside bend radius (>= min. radius) , die opening width W = 8 t for V, die depth so as to allow the part, clearance in U-die c = t + 0.05 - 0.1 mm (max thickness + allowance), side relief .
- Pressure pad / knock-out and stops (pins or gauge plate) to locate the blank ; pilot pins to prevent the slippage (blank skids when bending).
- Material: tool steel (T 105 / D2) hardened 55-60 HRC; finish polished (Ra 0.2 - 0.4).
- Bending sequence: bend the outside bends first , then inside bends; bend lines at right angle to the grain.

12. Drawing Dies

- Drawing: converts a flat blank into a hollow cup (cylindrical, rectangular or irregular) by pushing a punch into a die; the flange flows radially inward (the sheet is under radial tension and circumferential compression).
- Types: first draw (cupping), redraw (reverse and direct), ironing (thin the wall), deep drawing, stretch forming, hydro-forming ; single-acting, double-acting (blank holder) press.

12.1 Blank development

- Equal surface area principle (thickness unchanged): Area of blank = area of the finished cup (with trim allowance).
- Cup of diameter d and height h (flat bottom, sharp corner): D = sqrt(d2 + 4 d h) .
- With bottom corner radius r: D = sqrt(d2 + 4 d h - 0.5 r d) (approximate; r = bottom corner radius) add 2 - 5 mm trimming allowance on the height.

12.2 Draw ratio and number of draws

Drawing parameters

- Draw ratio DR = D / d (limit 1.8 - 2.0 for first draw for steel); reduction % = (D - d)/D x 100
- First draw: 40 - 50 % ; redraws: 25 - 30 % (second), 15 - 20 % (third) ... with intermediate annealing after 2 - 3 draws
- Drawing force: F = pi d t sigma_u x (D/d - 0.7) (empirically); or F = pi d t sigma_u x C C = 0.6 - 1.0
- Blank holder force: FBH = (1/3) x F or p x A : p = 1.5 - 3 N/mm2 (steel) x area of the flange under the holder
- Press force = F + FBH (+ margin 30 %); Work done = F x h x (0.6 to 0.7)

- Example: cup d = 50 mm, h = 40 mm, t = 1 mm: D = sqrt(2500 + 4 x 50 x 40) = sqrt(10 500) = 102.5 mm ; DR = 2.05 -> 2 draws needed ; first draw d1 = 0.6 x 102.5 = 61.5 mm (40 % reduction), second to d = 50 mm (19 % reduction).

12.3 Drawing die design

- Punch and die corner radii: die radius r_d = 4 to 10 t (soft/thin small, hard/thick large); punch radius r_p = (3 to 4) t ... up to 10 t (maximum = 1/2 of wall). Small die radius -> tearing (fracture) , large -> wrinkling.
- Clearance (per side) c = t + (0.07 to 0.2) x sqrt(10 t) or c = 1.1 t to 1.3 t (steel 1.1 t, aluminium 1.05-1.1 t, stainless 1.2 t); too tight -> ironing, tearing .
- Punch diameter = d (inside of cup) ; die diameter = d + 2t + 2 (c - t) i.e. punch + 2c.
- Blank holder (pressure pad) prevents wrinkles; spring, rubber, pneumatic or double-action press hold; draw beads control the flow.
- Air vent in the punch (hole 3-6 mm) to avoid vacuum when stripping the cup.
- Lubrication: heavy oil, wax, polyethylene film, soap; die material alloy steel / cast iron, chrome plated or nitrided ; aluminium bronze inserts for stainless steel; carbide for long-run; finish Ra 0.1 - 0.2 .
- Defects: wrinkling (flange: low blank holder force, big DR), tearing / fracture (too much force, small radius), earing (anisotropy), surface scratch , springback of walls.

13. Design and Drafting of Press Tools

13.1 Design procedure

1. Study the component drawing - material, thickness, tolerance, quantity, press available.
2. Decide the operations and the sequence ; select type of die .

3. Strip layout -> pitch, strip width, utilisation.
4. Calculate clearance, cutting force, stripping force, press tonnage , centre of pressure.
5. Determine die block, punch, stripper, die set dimensions; select standard die set (IS 12 ... or manufacturer) with guide pillars .
6. Select materials & heat treatment ; fasteners , springs, stops.
7. Check press data: shut height, bed and bolster size, stroke, shank diameter.
8. Prepare assembly and detailed drawings , BOM, and check .

13.2 Drafting practice (what to show)

- Plan (top view) with upper half removed: shows die block, stripper, guide rails, stops, strip layout in chain-line. Elevation: sectional front view showing punch, die, stripper, shoe, pillar, springs, shank in closed position.
- Strip layout and component drawing on the sheet; the die set is in 'open and closed' or shown in section at the 'closed position' .
- Bill of materials: part no., name, quantity, material, heat treatment, standard/remarks. Notes: hardness, tolerances (punch & die clearance given), grinding allowance, dowel/ screw positions , and the shut height .
- Standard die set parts: upper shoe, lower shoe, guide pillar (ground, press-fit H7/p6), guide bush (slip fit H7/h6 or ball-bearing bush), shank (centre of pressure), punch holder, back-up plate, stripper, spacers.

Exam checklist for any die design question

- Write operation, material, thickness, tau ; clearance , punch & die size ; cutting force , stripping , press tonnage
- Draw strip layout (pitch, margin, bridge, utilisation) and centre of pressure
- Sketch the die assembly (plan + section) with names of all parts; give the BOM

14. Quick Revision - Unit IV

- Dies: simple, compound, combination, progressive, transfer. Clearance c = 0.0032 t sqrt(tau); blank -> die size; hole -> punch size .
- F = L t tau; stripping 10 %; press tonnage with 30 % margin; shear on punch/die reduces force.
- Pilots register the strip; stripper (fixed/spring); pressure pad / knock-out.
- Centre of pressure = (sum L x)/(sum L); shank at CP. Strip layout: a = t to 1.5 t, utilisation = blank area/(W x P).

- Bending: BA = (pi/180) theta (R + k t); springback -> overbend. Drawing: D = sqrt(d2 + 4 d h); first draw 40-50 %, die radius 4-10 t, c = 1.1 t.

TOOL DESIGN

Subject Code : 23A03504a (Professional Elective - IV)

UNIT V : TOOL DESIGN FOR CNC MACHINE TOOLS

Syllabus (as given)

Introduction - Tooling requirements for Numerical control systems - Fixture design for CNC machine tools - Sub plate and tombstone fixtures - Universal fixtures - Cutting tools - Tool holding methods - Automatic tool changers and tool positioners - Tool presetting - General explanation of the Brown and Sharp machine.

1. Introduction

- *Numerical Control (NC):* machine tool movements and functions are controlled by coded numerical instructions (program) - G-codes (motion) and M-codes (miscellaneous).
- *CNC (Computer Numerical Control):* an on-board computer (controller) reads the part program, interpolates the paths and drives the servo motors of axes with feedback from encoders/ scales.
- *DNC:* many CNC machines fed from a central computer. CNC machining centre (MC) = milling machine + ATC + pallet changer; turning centre = lathe with turret and live tools.

Feature	Conventional machine	CNC machine
Control	operator's skill and hand wheels	program + servo drive
Accuracy	operator dependent	+/- 0.005 mm, repeatable
Set-up	jigs/fixtures needed for every job	simple universal fixtures , offsets in controller
Tool change	manual	automatic (ATC)
Flexibility	low	high - change the program
Tooling cost	high (special tools)	general-purpose tooling , modular

- *Axes:* Z is along the spindle; X and Y in table plane (milling); A, B, C rotary about X, Y, Z.

Coordinate systems: machine zero (home), work offset (G54-G59) , tool length offset (G43) , cutter compensation (G41/G42) .

- Consequence for tooling: CNC aims at low non-cutting time , few set-ups , machining many faces in one clamping -> fixtures and tools must be accurate, rigid, quick-change, and presettable .

2. Tooling Requirements for Numerical Control Systems

- Tools for CNC must be:

1. More rigid and accurate - because the machine runs at high speed and feed , with unattended operation.
2. Wear resistant with long and consistent life - carbide / coated / ceramic / CBN / PCD inserts ; because tool change interrupts production (tool life is planned , tool life management in the controller).
3. Standardised and quick-change - standard shank (BT, CAT, HSK, ISO) , standard tool length and diameter ; modular tooling system.
4. Presettable offline - length and diameter measured outside the machine -> offsets are entered (no machine downtime).
5. Good chip control - chip breakers , high-pressure coolant; since the operator isn't there to clear chips; chips must not wind round the tool .
6. Reliable and fool-proof - tool monitoring (breakage detection, tool load) .
7. Dimension stability - thermal stability of holders; balanced for high rpm (G2.5 at 15 000 rpm).
8. Minimum variety of tools - use multi-purpose tools so that ATC magazine capacity (20-100) suffices.

Main differences from conventional tooling

- Tool is located by the machine using offsets -> no set block / feeler gauge setting
- Work is located by fixture + probe ; datum known to the program
- Everything in tooling is predictable : the programmer decides cut conditions , so repeatability is vital
- Tooling economics: higher cost per tool but lower per part (long life, low scrap)

- Cutting tool materials for NC: HSS-Co, coated carbide (TiN, TiCN, TiAlN, Al2O3 CVD / PVD) , cermet, ceramic, CBN, PCD ; indexable inserts dominate (CNMG, DNMG, WNMG for turning, SDMT, APMT for milling).

3. Fixture Design for CNC Machine Tools

- A CNC fixture locates the work accurately relative to machine zero , clamps it rigidly against high forces , gives full access to the tool (multi-face machining) and allows quick changeover .

3.1 Requirements specific to CNC fixtures

- Zero (datum) point: the fixture has a known and repeatable datum (the fixed corner/ locator , a zero-point system) related to the machine origin ; the programmer uses it as work offset (G54 ...) . So fixture position must be fixed accurate and repeatable .
- Positive, accurate location by dowel pins and keys in table T-slots or grid holes , or zero-point clamping system (repeatability 0.005 mm).
- Full tool access: no clamps in the tool path (clamps low profile , pulled-back), tool can approach 5 faces in one set-up; check by simulation (collision).
- Rigidity: clamps near the cutting and over supports; vibration damping ; base thickness enough ; minimum overhang ; short tool length .
- Chip evacuation: open design, slots, inclined surfaces, high-pressure coolant and chips must not accumulate on locators ; through holes in the base .
- Modularity and standardisation: standard grid plates, tooling blocks, risers, toe clamps, 'T-nuts', hydraulic vices.
- Quick changeover: hydraulic / pneumatic clamping , pallets (work is loaded outside the machine while the machine cuts the previous pallet).
- Fool-proofing and verification: part presence sensors; probe checks; locators that prevent wrong loading.
- Light weight (for pallet-changers) , the weight limit of table/pallet load .

3.2 Types of clamping used

- Mechanical: strap and toe clamps, edge clamps (pull-down), step clamps , screw-type 'side' clamps .
- Hydraulic: swing clamps, work supports (retractable), hydraulic vices, pull-down clamps; high force, fast, uniform.
- Pneumatic: quick for light cuts ; vacuum fixture: for sheets and plates (large flat area, no distortion).
- Magnetic (permanent / electro) , low-melting alloy or wax embedding for delicate shapes (turbine blades).

3.3 Design procedure (CNC fixtures)

1. Study part & process plan: operations, face to be machined, datums, tolerance, cutting forces.
2. Choose machine and table layout (T-slot pitch, hole grid, pallet size, travel).
3. Select locating scheme (3-2-1) from the finished/ machined datum ; first operation on rough faces by adjustable pads .
4. Design clamping (positions, force) so that tool path is free .
5. Decide zero point and offsets ; probe cycle for alignment if required.
6. Check rigidity (deflection < 1/3 tolerance) , tool access (collision) , chip & coolant flow .
7. Draw, simulate with CAD/CAM, prove out by the first-off and revise .

- Fixture cost guidelines: keep to 10-20 % of the part cost over the production run ; use modular where lots small .

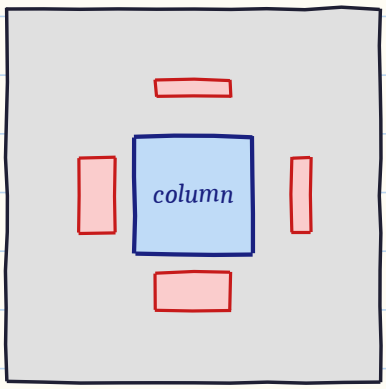
4. Sub-plate and Tombstone Fixtures

4.1 Sub-plate (grid plate) fixtures

- A sub-plate is a flat, thick, precisely ground plate (steel or cast iron) with a grid of accurately spaced tapped holes and dowel holes (e.g. 25 or 50 mm pitch, +/- 0.01 mm) that is bolted and dowelled to the machine table or pallet once and stays on the machine .
- Work-holding elements (clamps, locators, vices, risers, angle plates) are mounted anywhere on the grid with a fixed known position ; so the fixture is built quickly and repeated by recording coordinates.
- Advantages: quick set-ups, reusable, accurate and repeatable positions (known coordinates), easy to program (the position of every locator is known) , protects the machine table from damage; a standard for small-lot production .
- Types: plain grid (tapped + dowel holes), T-slot sub-plate, vacuum sub-plate, a sub-plate with a pallet system; tooling plate with a hole-pattern for zero-point clamping .
- Limitation: Limited surface, can't be used for machining all faces ; height is added -> reduces the travel .

4.2 Tombstone fixtures

Top view : 4 faces, 4 parts



Tombstone fixture

Tall block (cube / column) mounted on a pallet or rotary table of a horizontal machining centre (HMC).
Parts clamped on 2, 3 or 4 faces.
Table indexes 90 deg -> next face cuts without stopping -> less idle time.
Used with pallet changer: load one tombstone while the other is machining.

- A tombstone is a tall rectangular (or square) column with work-holding on its 2, 3 or 4 vertical sides , fixed on a pallet or the rotary table of a horizontal machining centre (HMC) .
- Operation: machine the part on one face , index the table (B-axis) 90 deg (or 180), machine the next face... each face can hold one or many (many identical parts, or several different parts, 'family' of parts) ; the operator loads the faces on the other pallet while the machine is cutting .
- Advantages: high spindle utilisation (low idle time), more parts per set-up, more tools used on the single loading, multiple faces by one fixture, accurate relationships between faces (single set-up), fewer pallets for pallet pool , good chip fall (vertical faces) .
- Design points: tombstone is heavy and rigid ; keep the centre of gravity low and over the table centre ; height within the Y-travel and tool reach ; chip fall-through gaps ; strong (grid / dowel hole) faces ; face-to-face accuracy : faces ground parallel/ perpendicular within 0.01 mm ; coolant and hydraulics through-the pallet for the hydraulic clamps (rotary union on the table) .
- Variants: cube (4 faces), 'dual-face' (2 sides), 'triangular' ; cast iron or welded steel/ aluminium , with a grid or T-slots on each face.

4.3 Comparison

Item	Sub-plate (grid plate)	Tombstone
Position on machine	flat on table / pallet	vertical column on pallet (HMC)
Faces machined	1 (top) + sides with angle plate	2 to 4 sides, many parts

Item	Sub-plate (grid plate)	Tombstone
Use	VMC / small lot, many different jobs	HMC, medium to high volume
Set-up time	short	very short per part (many parts/set-up)
Main advantage	flexibility, known grid	spindle utilisation, accuracy of faces

5. Universal Fixtures

- Universal (general-purpose / flexible) fixtures: standard work-holding that can be used for a wide range of different parts with little or no modification - the fixture is not designed for one part .
- Examples: machine vices (with soft jaws), chucks (3-jaw, 4-jaw, collet, magnetic), rotary tables and index heads, angle plates, V-blocks, parallels, step blocks and clamps, swivel / tilting tables (trunnion), vacuum tables, tooling-plate kits, modular fixturing systems.
- Modular fixture systems: a set of standard hardened and ground elements (base plates with T-slot or hole grid, riser blocks, locators, supports, clamps) are assembled into a fixture for a job, used, then dismantled and kept for reuse .
  - Types: T-slot based modular system dowel-pin / hole-based system (grid; '16 mm hole, 50 mm pitch' common), and Combi-/ phase systems.
  - Advantages: low lead time (hours), reuse of components, lower cost for small lots, accurate (precision elements), good for prototypes, flexible manufacturing systems (FMS) .
  - Limitations: less rigid (more joints), initial cost, assembly skill needed, and bigger size than a dedicated fixture .
- Other universal approaches: hydraulic / zero-point clamping on all vices and plates , adjustable (reconfigurable) pin-bed / 'bed-of-nails' fixtures , phase-change (low-melting alloy, ice) embedding , and programmable / adaptive fixtures with robot .
- Group technology (GT): the parts of a family of similar shape are held in one adjustable fixture (' group fixtures '), by changing the inserts, jaws or locators only.

Fixture type vs production volume

- Low volume, many variants -> universal / modular / sub-plate
- Medium volume, families -> tombstone, group (adaptable) fixtures
- High volume, single part -> dedicated hydraulic fixture / pallet

6. Cutting Tools for CNC

- Types of cutting tools used on machining centres: face mills, end mills (solid carbide / indexable), ball-nose and bull-nose cutters, slot drills, shell mills, T-slot and thread mills, drills (twist, indexable, through-coolant, spade), reamers, boring heads (fine adjustable), taps (rigid tapping), chamfer and counter-sink tools, form tools .
- Turning centre tools: ISO (indexable) turning inserts and holders (external, internal, grooving, parting off, threading), boring bars (damped), drills , live (driven) tools for milling on the lathe.

6.1 Indexable insert tools

- Inserts clamped on the tool body by screw, top-clamp, lever, wedge or clamp-pin ; edge indexed to a fresh edge when worn - no regrinding; consistent size (+/- 0.01) . Several shapes (C, D, R, S, T, V, W) , clearance angles (N, B, C ..) , tolerance classes (G, M, U) , chip-breaker geometry (Seen in ISO 1832 code : e.g. CNMG 120408 = C shape 80 deg, 0 deg clearance (N), tolerance M, hole with chip-breaker (G), size 12 mm, thickness 04, nose radius 0.8 mm).
- Selection factors: work material (ISO P - steel (blue), M - stainless (yellow), K - cast iron (red), N - non-ferrous (green), S - superalloy (brown), H - hardened (grey)) ; operation (roughing, finishing) ; machine power & rigidity ; tool life & cost .
- Tool geometry for NC: generally positive-rake inserts for finishing and low power , negative-rake (strong, double-sided so with 2 x the edges) for heavy roughing .

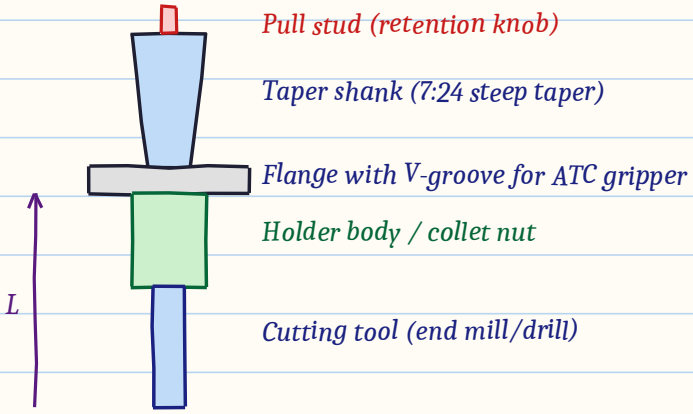
6.2 Tool selection rules for CNC

1. Choose largest suitable tool diameter and the shortest possible stick-out (overhang) for rigidity (deflection ~ L3/D4).
2. Use standard tool sizes and 'one tool, many jobs' (minimising the tool changes).
3. Use through-spindle coolant tools for deep holes and chip evacuation.
4. For finishing use wiper inserts and light cut ; for roughing use coarse pitch cutters with chip splitters .
5. Tool life management: life in minutes or parts counted by controller, sister tools

(duplicate) used automatically when life is over.

6. Spindle speed N = 1000 V/(pi D) ; feed rate F = fz x z x N ; cutting data from catalogue .

7. Tool Holding Methods



Tool presetting data

Tool length L (gauge line to tip)
Tool diameter D (radius offset)
Measured OFF the machine on a presetter -> values entered in the tool offset table of the CNC

- Tool holder: a standard interface between the machine spindle and the cutting tool ; it must provide accurate location (concentric, low runout < 0.005 - 0.01 mm), high clamping force, high stiffness, balance, and be compatible with the ATC (gripper groove + pull stud) .

7.1 Spindle (shank) interfaces

- 7:24 steep taper (self-releasing; ISO / BT / CAT / DIN 69871 (SK) 30, 40, 50) - most common on machining centres , the taper is pulled into the spindle by a drawbar and pull-stud (retention knob) ; flange is gripped by ATC.
- BT (Japanese; symmetric - good for high speed), CAT (American V-flange), SK/DIN (German), ISO.
- HSK (hollow short taper 1:10): hollow taper shank + face contact (dual contact) -> high accuracy/ stiffness at high rpm (above 20 000 rpm) e.g. HSK-A63; Capto (polygon taper PSC) for turning/ milling.
- Morse taper (MT 1 - 6), R8, 'Weldon' flats on older / drilling machines.
- Turning: VDI (DIN 69880) and BMT (base mount) turret tool holders , square shank (25 x 25) holders , boring bars clamped in bushes , quick-change tool posts .

7.3 Methods of holding the cutting tool in the holder

Method	How	Remarks

Method	How	Remarks
Collet chuck (ER collet)	nut closes a slotted collet around the shank	versatile, runout 0.01 mm , common; many sizes
Side-lock (Weldon) holder	screw presses on a flat on the tool	high torque, no pull-out; poor concentricity - roughing
Shrink-fit holder	holder bore heated (induction 300 C), tool put in; shrinks on cooling	run out < 0.003 mm, perfect balance, high stiffness , high rpm; need heater
Hydraulic chuck	hydraulic pressure expands a thin sleeve	quick, vibration damping, 0.003 mm accuracy
Milling chuck (high-torque)	needle roller bearings force a sleeve	high grip for heavy roughing
Face-mill arbor (shell mill holder)	tool bolted on the face, driven by a key	big face mills
Drill chuck / Jacobs, tapping holder	keyless drill chuck; compression-tension tapping head	drilling, tapping
Boring head	micro adjustable slide	bore sizes with 0.002 mm adj.
Modular / quick-change	tool head screwed or taper-locked onto the shank	reduces tool inventory

- Balancing: unbalanced holder -> vibration at high rpm; G2.5 or G6.3 balancing grade ; symmetric design , nut with balance screws .

8. Automatic Tool Changers and Tool Positioners

8.1 Automatic tool changer (ATC)

- ATC swaps the tool in the spindle for the next one automatically on the program command (M06, T-code) within 1 - 10 seconds (chip-to-chip 2-6 s) , which makes the multi-operation machining possible without an operator .
- Main parts: tool magazine (storage) , tool changer arm (gripper) / mechanism , tool-selection (positioning) system , spindle with drawbar and clamp (spring clamp + hydraulic unclamp) .
- Types of magazines:

1. Drum (disc) magazine: tools in the radial / axial pockets of a rotating disc; capacity 12 - 40 ; the whole magazine moves to spindle (shuttle) ; simple, fast .
2. Chain magazine: tools on a chain with sprockets ; capacity 40 - 120+ ; (horizontal, vertical or oval) ; big and flexible .
3. Rack (matrix) / bin magazine: tools in rows , accessible by robot / gantry ; capacity 100 - 500 ; used in large machines & FMS.
4. Turret (lathe) : 8 - 12 stations indexed (servo or Hirth coupling), the tool changing = turret indexing in 0.2 - 1 s .

- Changing methods: (a) Arm type (double-arm, 180 deg swing) : gripper grabs the new and old tools simultaneous - fastest; (b) Direct spindle-to-magazine ('pick-up') : spindle moves to the magazine; no arm - cheaper, slower; (c) Robot / gantry type .
- Tool selection: Sequential (tools in the order of use; simple), random (memory) selection - the controller keeps track of the tool's pocket; the tool is put wherever the pocket is free; 'fixed' pocket addressing with tool code (bar code, RFID chip).

ATC sequence (random / arm type)

- Magazine indexes the next tool to standby position (parallel to cutting)
- On M06: spindle goes to the tool change position (Z home), orients (M19)
- Arm swings: grips both tools -> unclamps spindle -> pulls out -> rotates 180 deg -> inserts -> clamps
- Arm returns; spindle goes back and magazine places the old tool in the pocket

8.2 Tool positioners

- Tool positioner: a device that accurately brings the tool into a defined (known) position or indexes tool holders , e.g. the lathe turret , the tool-setter (probe on machine) , a tool pre-setter or the magazine indexing mechanism .
- Features: Hirth (serrated) coupling in turret for repeatability 0.002 mm; Geneva mechanism / servo with encoder for magazine indexing ; keys for tool orientation (the drive keys on the flange, spindle orientation M19) .
- Tool offsets (length and radius) compensate for the differences in tool position (so the program uses nominal values).
- Tool monitoring / tool breakage detection : touch probe or laser, spindle-load monitoring, acoustic emission , tool-life counters .

9. Tool Presetting

- Tool presetting = measuring the length and diameter of each tool (with its holder) outside the machine in a tool presetting device , and entering the values as tool offset in the controller, so the machine is not stopped for tool setting .
- Presetter: a precision spindle with the same taper (e.g. BT40) mounted on a column , with optical (shadow-graph/ camera) / touch-probe measuring unit and a digital display ; data transfer by RFID chip / data cable / DNC to the machine .
- What is measured: tool length L from the gauge line (flange) to the cutting tip, the radius/diameter D, the nose radius, run-out, and profile check .

Benefits of presetting

- Machine idle time is reduced (no on-machine setting)
- Tool life management and consistent first-part accuracy
- Detects tool defects, wrong tool or runout before use
- Preset data stored with tool ID (RFID), avoid human errors in offset entry

- Methods: (1) off-line presetting (optical / contact) , (2) on-machine tool setters (touch probe / laser) - measure automatically during the program and update wear offset , (3) manual touching of reference block / 'touch-off' (slow but no equipment) .
- Offsets used: G43 H.. = tool length offset ; G41/G42 D.. = cutter radius compensation ; wear offsets are modified after first part inspection .
- Typical accuracy of presetters: length +/- 0.005 mm; diameter +/- 0.003 mm ; temperature controlled room and calibrated masters .

10. General Explanation of the Brown and Sharp Machine

- Brown & Sharpe is a famous American machine-tool and precision-instrument maker (David Brown, joined by Lucian Sharpe; the name Brown & Sharpe dates from 1853). It is known for the universal milling machine (1862), the micrometer (1867), grinders, measuring machines (later CMMs) and the automatic screw machine .
- In this unit the Brown & Sharpe machine is explained as the classic cam-controlled automatic screw machine : bar stock is fed through a hollow spindle, held by a collet, and cut by tools on cross slides and a turret, all movements controlled by cams - the mechanical forerunner of CNC for mass producing small parts (screws, pins, bushes).
- (Exam note: also quote your prescribed textbook's description of the Brown & Sharpe machine and its control system if your faculty has stressed it.)

10.1 Main parts

- Headstock with hollow spindle and collet chuck (feed and clamp by cams) , bar stock magazine and feed finger .
- Cross slides (front, rear, vertical) carrying form tools, cut-off tool ; turret (hexagonal, horizontal axis or sliding) with 5-6 stations for drills, reamers, dies, taps, box-tools.
- Camshaft (main cam shaft) with cam drums / plates which control (a) the slides, (b) the turret feed and index, (c) the collet opening & closing and bar feed, (d) speed change ; gear train for speeds and feeds.
- Stock stops, work holding on collet, coolant system.

10.2 Working and tooling

- Working cycle (one revolution of the cam shaft = one component): feed bar to the stop -> close the collet -> turret indexes to station 1 -> form tool/ drill/ turn/ thread -> cut off -> open collet . Cycle time set by cam speed change gear .
- Tooling: form tools (circular / dovetail) on the cross slide for contours , box tool for turning on turret, knee tools , drills & reamers , taps & dies (self-opening die heads), knurls , cut-off blade .
- Set-up: cam design (lobes with rise and fall give feed rate) , tool layout chart (sequence of tools at each station and cam) , form tool design (relief angle 10-15 deg & rake 0-15 deg).
- Advantages: full automatic (one operator for many machines) , very high production , good accuracy for small components ; tooling is simple. Limitations: long set-up time (cams), limited to bar stock, not flexible; replaced by CNC 'Swiss-type' lathes and turning centres.
- Relevance to NC: CNC replaces mechanical cams by program and servo drives - this removes the cam changing and gives flexibility ; the same principles of tool layout, tool holding, and tool-life management still apply .

11. Quick Revision - Unit V

- NC tooling: rigid, quick-change, presettable, long life, good chip control; standard shanks (BT, CAT, HSK).
- CNC fixtures: known zero point, full access, rigid, quick change; sub-plate = grid plate; tombstone = vertical multi-face on HMC; universal / modular = reusable standard elements .

- Tool holding: 7:24 taper + pull-stud, HSK; collet (ER), shrink-fit, hydraulic, side-lock.
- ATC: magazine (drum, chain, rack) + arm; random selection with memory; turret on lathes.
- Tool presetting: measure L and D offline -> offsets G43 / G41; Brown & Sharp: cam-operated automatic screw machine (pre-CNC automation).